AP CALCULUS BC · EXAMINATION REPOSITORY

Unit 9

Parametric Equations, Polar Coordinates, and Vector-Valued Functions

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings · Exactly 4 Questions Per Page)
- Questions 91–100: Free Response with Official AP Scoring Rubrics (9 Pts Each · 1 Per Page)
- Integrated AP Polar Roses, Cardioids, Limacons, Vector Motion Tables & SVG Graphs
- Complete 90–Question Answer Key & Step-by-Step Bulleted Solutions with Distractor Analysis

OFFICIAL AP CED ALIGNED · MEDIUM LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This comprehensive paper contains 100 continuously numbered questions strictly covering Unit 9 (Parametric Equations, Polar Coordinates, and Vector-Valued Functions), incorporating official AP-style vector trajectories, polar curve graphs, tables of values, and multi-part FRQ rubrics to test conceptual and analytical calculus mastery.

⭐ Priority Rating System

★★★★★ VERY IMPORTANT	High-frequency core exam topics: Parametric First & Second Derivatives (dy/dx and d^2y/dx^2), Planar Vector Particle Motion (Velocity, Acceleration, Speed), Area Enclosed by Single & Double Polar Curves, and Polar Tangent Slopes.
★★★★ MANDATORY	Essential analytical skills: Parametric & Polar Arc Length, Particle Collisions vs. Intersecting Paths, Rate of Change of Polar Area/Radius, Horizontal & Vertical Tangent Lines, and Vector Antidifferentiation with Initial Conditions.
★★★ IMPORTANT	Foundational practice: Polar to Cartesian conversions, speed increasing/decreasing determination, parametric concavity, and single-petal rose area evaluations.

- **Questions 1–90:** Multiple Choice, four choices, exactly one correct answer (Exactly 4 MCQs per page).
- **Questions 91–100:** Free Response Questions, 9 points each, with authentic AP multi-part rubrics (1 FRQ per page).
- **Visual Elements:** Crisp vector SVG polar diagrams, particle motion trajectories, velocity field representations, and numerical step tables.
- **Calculator Policy:** No calculator required (Closed-form analytical evaluation).
- **Answer Section:** Complete Answer Key grid, followed by step-by-step bulleted solutions and distractor breakdowns.

Topic Coverage (Unit 9.1 – 9.9)

9.1 Defining & Differentiating Parametric Equations (dy/dx, Tangents)	9.2 Second Derivatives of Parametric Equations (d^2y/dx^2, Concavity)
9.3 Arc Lengths of Curves Given by Parametric Equations	9.4 Defining & Differentiating Vector-Valued Functions (Velocity, Speed)
9.5 Integrating Vector-Valued Functions & Initial Value Problems	9.6 Solving Motion Problems (Collisions vs. Crossing Paths)
9.7 Defining Polar Coordinates & Differentiating in Polar Form (dy/dx)	9.8 Finding the Area of a Polar Region Bounded by a Single Curve
9.9 Finding Area Bounded by Two Polar Curves & Polar Arc Length	9.10 Synthesis: Motion Along Polar & Parametric Paths

Essential Tip: When evaluating parametric second derivatives, remember the quotient formula: $d^2y/dx^2 = [\, d/dt(dy/dx) \,] / [\, dx/dt \,]$. Never compute $(d^2y/dt^2) / (d^2x/dt^2)$! In polar coordinates, always convert $x = r\cos\theta$ and $y = r\sin\theta$ before computing Cartesian tangent slopes dy/dx.

ITEM 01

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is defined by the parametric equations $x(t) = t^3 - 3t$ and $y(t) = t^2 + 2$ for all real numbers t. What is the slope of the line tangent to the curve at the point where $t = 2$?

A 4 / 9
B 9 / 4
C 4 / 3
D 2 / 3

ITEM 02

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve in the xy-plane is given by the parametric equations $x = e^{2t}$ and $y = \sin(3t)$. What is the equation of the line tangent to the curve at the point corresponding to $t = 0$?

A $y = (2/3)(x - 1)$
B $y = (3/2)(x - 1)$
C $y = 3(x - 1)$
D $y = (3/2)x$

ITEM 03

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is defined by $x(t) = t^3 - 12t$ and $y(t) = 2t^2 - 8$ for all real numbers t. At which value(s) of t does the curve have a horizontal tangent line?

A $t = 0$ and $t = \pm 2$
B $t = \pm 2$ only
C $t = 0$ only
D No values of t

ITEM 04

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A curve is given by $x(t) = t^2 - 4t$ and $y(t) = t^3 - 6t$ for all t. At which value(s) of t does the curve have a vertical tangent line?

A $t = 2$ and $t = \pm\sqrt{2}$
B $t = \pm\sqrt{2}$ only
C $t = 0$ and $t = 4$
D $t = 2$ only

ITEM 05

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with parametric equations $x(t) = 4\cos t$ and $y(t) = 3\sin t$ for $0 \le t \le 2\pi$. What is the slope of the curve at $t = \pi / 4$?

A $-3/4$ **B** $3/4$

C $-4/3$ **D** $4/3$

ITEM 06

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve C is defined by $x(t) = t - \ln t$ and $y(t) = t + \ln t$ for $t > 0$. For what value of t is the line tangent to C parallel to the line $2x - y = 5$?

A 2 **B** 3

C 1 **D** $1/3$

ITEM 07

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The table below gives values of x(t), y(t), x'(t), and y'(t) for selected values of t.

t	x(t)	y(t)	x'(t)	y'(t)
1	3	−2	4	−8
2	5	1	−3	6
3	2	7	2	−1

What is the equation of the line tangent to the parametric curve at the point where $t = 2$?

A $y - 1 = -(1/2)(x - 5)$ **B** $y - 1 = 2(x - 5)$

C $y - 1 = -2(x - 5)$ **D** $y + 2 = -2(x - 3)$

ITEM 08

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

At what point (x, y) on the curve defined by $x = t^2 + 1$ and $y = t^3 - 3t$ does the tangent line have a slope of 0?

A (0, 0) **B** (1, 0)

C (2, 0) **D** (2, −2) and (2, 2)

ITEM 09

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A curve is given parametrically by $x(\theta) = 5\sec\theta$ and $y(\theta) = 5\tan\theta$ for $-\pi/2 < \theta < \pi/2$. What is dy/dx in terms of θ?

A $\csc\theta$ **B** $\sin\theta$

C $\cos\theta$ **D** $\sec\theta$

ITEM 10

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is defined by $x = t^3 - 3t^2$ and $y = t^3 - 3t$. At how many distinct points on the curve does the tangent line have a slope of 1?

A 2 **B** 1

C 3 **D** 0

ITEM 11

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is defined by the parametric equations $x(t) = t^2$ and $y(t) = t^3 - 3t$ for $t > 0$. What is the value of d^2y / dx^2 at the point where $t = 1$?

A $3/4$ **B** 3

C $3/2$ **D** 6

ITEM 12

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is given by $x(t) = \cos t$ and $y(t) = \sin t$ for $0 < t < \pi$. What is the value of d^2y / dx^2 at $t = \pi / 6$?

A -1 **B** -4

C 8 **D** -8

ITEM 13

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A curve in the xy-plane is defined by $x = t^3 + 1$ and $y = t^2 - 2t$ for $t > 0$. What is d^2y / dx^2 at $t = 1$?

A 2 / 9
B −2 / 9
C 2 / 3
D 1 / 3

ITEM 14

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is given by $x(t) = e^t$ and $y(t) = t^2$ for all real numbers t. On which interval of t is the curve concave downward ($d^2y / dx^2 < 0$)?

A $t < 1$
B $t > 1$
C $t > 0$
D $t < 0$

ITEM 15

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A curve is given by $x(t) = 3t$ and $y(t) = t^3 - 6t^2 + 9t$ for all real numbers t. At which point (x, y) does the curve have a point of inflection?

A (3, 4)
B (2, 2)
C (6, 2)
D (6, 0)

ITEM 16

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A student incorrectly claims that the second derivative of a parametric curve can be found using the formula $(d^2y/dt^2) / (d^2x/dt^2)$. For the curve $x(t) = t^2 + 1$ and $y(t) = t^4$, what is the TRUE value of d^2y / dx^2 at $t = 2$?

A 8
B 24
C 6
D 2

ITEM 17

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The functions x(t) and y(t) are twice-differentiable. At t = 3, values of the functions and their derivatives are given below:

t	x(t)	y(t)	x'(t)	y'(t)	x''(t)	y''(t)
3	4	1	2	6	−1	4

What is the value of d^2y / dx^2 at t = 3?

A 7 / 4
B 2
C 7 / 2
D 14

ITEM 18

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A curve is defined by x = ln t and y = √t for t > 0. What is d^2y / dx^2 at t = 4?

A 1 / 4
B 1 / 2
C 1 / 8
D 1

ITEM 19

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A cycloid is given by x(t) = t − sin t and y(t) = 1 − cos t for 0 < t < 2π. What is d^2y / dx^2 at t = π / 2?

A −1 / 2
B 1
C −1
D 0

ITEM 20

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A curve is defined by x(t) = $2t^3$ and y(t) = $3t^4$ for t > 0. What is d^2y / dx^2 at t = 1?

A 1 / 6
B 2 / 3
C 2
D 1 / 3

ITEM 21

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve given by the parametric equations $x(t) = \cos t + t \sin t$ and $y(t) = \sin t - t \cos t$ on the interval $0 \le t \le 3$?

A $9/2$
B 9
C 3
D 6

ITEM 22

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the spiral given by $x(t) = e^t \cos t$ and $y(t) = e^t \sin t$ from $t = 0$ to $t = \pi$?

A $e^{\pi} - 1$
B $\sqrt{2}(e^{\pi} - 1)$
C $2(e^{\pi} - 1)$
D $\sqrt{2}\, e^{\pi}$

ITEM 23

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the astroid curve given by $x(t) = \cos^3 t$ and $y(t) = \sin^3 t$ in the first quadrant $(0 \le t \le \pi/2)$?

A $3/4$
B 3
C $3/2$
D 6

ITEM 24

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following definite integrals represents the total length of the parametric curve $x(t) = t^3 - 2t$, $y(t) = 4t^2 + 1$ for $1 \le t \le 5$?

A $\int_1^5 \sqrt{((t^3 - 2t)^2 + (4t^2 + 1)^2)}\, dt$
B $\int_1^5 \sqrt{((3t^2 - 2)^2 + 16t^2)}\, dt$
C $\int_1^5 ((3t^2 - 2) + 8t)\, dt$
D $\int_1^5 \sqrt{((3t^2 - 2)^2 + 64t^2)}\, dt$

ITEM 25

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the length of the curve $x(t) = 3t^2$ and $y(t) = 2t^3$ from $t = 0$ to $t = \sqrt{3}$?

A 14 **B** 16

C 7 **D** 28

ITEM 26

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

What is the distance traveled by a particle moving along the curve $x(t) = 2 \sin t$, $y(t) = 2 \cos t$ from $t = 0$ to $t = 3\pi / 2$?

A 6π **B** 3π

C $3\pi / 2$ **D** 4π

ITEM 27

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle travels in the xy-plane with velocity components $x'(t) = t^2 - 1$ and $y'(t) = 2t$ for $t \geq 0$. What is the total distance traveled by the particle from $t = 0$ to $t = 3$?

A 6 **B** 9

C 12 **D** 15

ITEM 28

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The velocity components of a particle moving in the xy-plane are recorded at selected times in the table below.

t	0	1	2	3
x'(t)	3	4	0	6
y'(t)	4	3	5	8

Using a trapezoidal sum with the three subintervals indicated by the table, what is the approximation for the distance traveled by the particle from $t = 0$ to $t = 3$?

A 20.0 **B** 35.0

C 15.0 **D** 17.5

ITEM 29

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve defined by $x(t) = \ln(\sec t)$ and $y(t) = t$ for $0 \le t \le \pi / 4$?

A $\ln(\sqrt{2} + 1)$
B $\ln(\sqrt{2} - 1)$
C $\sqrt{2} + 1$
D $\ln(2 + \sqrt{3})$

ITEM 30

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A particle travels along the path $x(t) = 6t$ and $y(t) = 8t$ for $t \ge 0$. The length of the path traversed by the particle from $t = 0$ to $t = k$ is 50. What is the value of k?

A 10
B 5
C 25
D 50

ITEM 31

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with position vector $r(t) = \langle t^3 - 3t, 2t^2 \rangle$ for $t \ge 0$. What is the acceleration vector of the particle at time $t = 2$?

A $\langle 6, 4 \rangle$
B $\langle 9, 8 \rangle$
C $\langle 12, 4 \rangle$
D $\langle 12, 8 \rangle$

ITEM 32

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

The position of a particle in the xy-plane is given by $r(t) = \langle 4\cos(2t), 4\sin(2t) \rangle$ for $t \ge 0$. What is the speed of the particle at any time t?

A $8\cos(2t)$
B 4
C 16
D 8

ITEM 33

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The position of a particle moving in the xy-plane is given by $r(t) = \langle e^{2t}, \ln(t+1) \rangle$ for $t \geq 0$. What is the speed of the particle at time $t = 0$?

A $\sqrt{5}$ **B** 3

C $\sqrt{3}$ **D** 5

ITEM 34

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with position vector $r(t) = \langle t^2 - 6t, 2t^2 \rangle$ for $t \geq 0$. At what time t are the particle's velocity vector and acceleration vector orthogonal (perpendicular)?

A 1.5 **B** 0.6

C 3.0 **D** 1.2

ITEM 35

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with velocity vector $v(t) = \langle 2, 2\sqrt{3} \rangle$ for all $t \geq 0$. What is the direction angle θ of motion of the particle, measured counterclockwise from the positive x-axis?

A $2\pi / 3$ **B** $\pi / 6$

C $\pi / 3$ **D** $\pi / 4$

ITEM 36

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with position vector $r(t) = \langle t^2 - 2t, 3t \rangle$ for $t \geq 0$. On what interval of t is the speed of the particle increasing?

A $t > 0$ **B** $t < 1$

C $t > 2$ **D** $t > 1$

ITEM 37

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with twice-differentiable position coordinates. At time $t = 4$, the particle has velocity vector $v(4) = \langle -3, 4 \rangle$ and acceleration vector $a(4) = \langle 2, -1 \rangle$. At time $t = 4$, is the speed of the particle increasing, decreasing, or neither?

A Decreasing, because $v(4) \cdot a(4) < 0$

B Increasing, because $v(4) \cdot a(4) > 0$

C Neither, because $v(4) \cdot a(4) = 0$

D Increasing, because $||v(4)|| = 5 > 0$

ITEM 38

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with velocity vector $v(t) = \langle 3t^2 - 12, 2t - 4 \rangle$ for all real numbers t. At what value(s) of t is the particle at rest?

A $t = \pm 2$

B $t = 2$ only

C $t = -2$ only

D The particle is never at rest

ITEM 39

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

The position of a particle is given by $r(t) = \langle \sin(3t), \cos(3t) \rangle$ for $t \geq 0$. What is the magnitude of the acceleration vector of the particle at any time t?

A 1

B 3

C 9

D 27

ITEM 40

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves along a path with position vector $r(t) = \langle t^2, 2t \rangle$ for $t > 0$. What is the unit tangent vector $T(t)$ in the direction of motion at time $t = 1$?

A $\langle 1, 1 \rangle$

B $\langle 2, 2 \rangle$

C $\langle 1/2, 1/2 \rangle$

D $\langle \sqrt{2}/2, \sqrt{2}/2 \rangle$

ITEM 41

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with velocity vector $v(t) = \langle 3t^2, 4t \rangle$ for $t \geq 0$. If the position of the particle at time $t = 1$ is $r(1) = \langle 2, 5 \rangle$, what is its position vector $r(3)$ at time $t = 3$?

A $\langle 28, 21 \rangle$

B $\langle 26, 16 \rangle$

C $\langle 27, 18 \rangle$

D $\langle 30, 26 \rangle$

ITEM 42

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

The acceleration vector of a particle is given by $a(t) = \langle 2, 6t \rangle$ for $t \geq 0$. If the particle's initial velocity is $v(0) = \langle 3, 1 \rangle$, what is its velocity vector at time $t = 2$?

A $\langle 4, 12 \rangle$

B $\langle 7, 13 \rangle$

C $\langle 7, 25 \rangle$

D $\langle 10, 14 \rangle$

ITEM 43

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A particle travels in the xy-plane with velocity vector $v(t) = \langle \cos t, \sin t \rangle$ for $0 \leq t \leq \pi$. What is the displacement vector of the particle over the interval $[0, \pi]$?

A $\langle -2, 0 \rangle$

B $\langle 0, 0 \rangle$

C $\langle 0, 2 \rangle$

D $\langle 0, 1 \rangle$

ITEM 44

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves with velocity vector $v(t) = \langle e^t, 2t \rangle$ for $t \geq 0$. At $t = 0$, the particle's position is $r(0) = \langle 1, -3 \rangle$. What is the position vector of the particle at time $t = \ln 3$?

A $\langle 4, -3 + 2\ln 3 \rangle$

B $\langle 2, -3 + (\ln 3)^2 \rangle$

C $\langle 3, (\ln 3)^2 \rangle$

D $\langle 3, -3 + (\ln 3)^2 \rangle$

ITEM 45

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle starts at the origin $r(0) = \langle 0, 0 \rangle$ and moves with velocity vector $v(t) = \langle 1/(t+1), 3t^2 \rangle$ for $t \geq 0$. At what time t is the particle's x-coordinate equal to $\ln 5$?

A 4
B 5
C $e^5 - 1$
D $\ln 4$

ITEM 46

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A projectile moves with acceleration vector $a(t) = \langle 0, -32 \rangle$ ft/s^2 for $t \geq 0$. Its initial velocity is $v(0) = \langle 60, 64 \rangle$ ft/s and its initial position is $r(0) = \langle 0, 80 \rangle$ ft. What is the maximum height achieved by the projectile?

A 64 ft
B 144 ft
C 128 ft
D 160 ft

ITEM 47

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The velocity vector of a particle moving in the xy-plane is given in the table below at selected times.

t	0	1	2	3
x'(t)	2	5	7	4
y'(t)	1	3	4	2

If the particle is at $r(0) = \langle 4, -1 \rangle$ at time $t = 0$, use a trapezoidal sum with three subintervals to approximate the position vector $r(3)$.

A $\langle 18, 6.5 \rangle$
B $\langle 15, 8.5 \rangle$
C $\langle 19, 7.5 \rangle$
D $\langle 19, 8.5 \rangle$

ITEM 48

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves with velocity vector $v(t) = \langle 2t, 1/\sqrt{t} \rangle$ for $t \geq 1$. If $r(1) = \langle 3, 4 \rangle$, what is its position vector $r(4)$ at time $t = 4$?

A $\langle 16, 6 \rangle$
B $\langle 15, 2 \rangle$
C $\langle 18, 8 \rangle$
D $\langle 18, 6 \rangle$

ITEM 49

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A particle starts at the origin $r(0) = \langle 0, 0 \rangle$ and travels with velocity vector $v(t) = \langle 6t, 8t \rangle$ for $t \geq 0$. What is the distance of the particle from the origin at time $t = 2$?

A 20
B 10
C 40
D 28

ITEM 50

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves with velocity vector $v(t) = \langle \sin(\pi t), \cos(\pi t) \rangle$ for $t \geq 0$. If the particle starts at the origin $r(0) = \langle 0, 0 \rangle$, what is its position vector at time $t = 2$?

A $\langle 2/\pi, 0 \rangle$
B $\langle 0, 0 \rangle$
C $\langle 0, 2/\pi \rangle$
D $\langle 1/\pi, 1/\pi \rangle$

ITEM 51

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Two particles move in the xy-plane for $t \geq 0$. The position of Particle A is given by $r_A(t) = \langle 2t + 1, 3t - 2 \rangle$ and the position of Particle B is given by $r_B(t) = \langle t^2 - 2, 2t + 2 \rangle$. Do the two particles collide, do their paths intersect without colliding, or neither?

A The particles collide at $t = 3$
B The particles collide at $t = 4$
C Their paths intersect, but the particles do not collide
D Their paths never intersect

ITEM 52

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with velocity vector $v(t) = \langle \cos(t^2), 3t \rangle$ for $t > 0$. At what time t does the particle first move purely vertically?

A $\sqrt{(3\pi / 2)}$
B $\sqrt{\pi}$
C $\pi / 2$
D $\sqrt{(\pi / 2)}$

ITEM 53

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle travels in the plane with position vector $r(t) = \langle t - 2, 2t - 1 \rangle$ for $t \geq 0$. What is the minimum distance from the particle to the origin?

A $3\sqrt{5}/5$

B $\sqrt{5}$

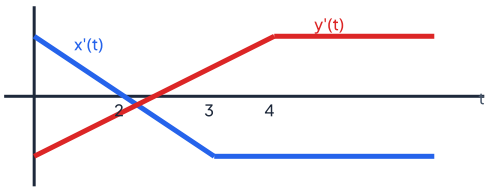
C $3/5$

D $2\sqrt{5}/5$

ITEM 54

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The graph below shows the velocity components $x'(t)$ and $y'(t)$ of a particle moving in the xy-plane for $0 \leq t \leq 6$.



During which time interval is the particle moving to the left and upward?

A $0 < t < 2$

B $3 < t < 6$

C $2 < t < 4$

D $0 < t < 3$

ITEM 55

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves with position vector $r(t) = \langle t^2, t^3 - 12t \rangle$ for $t \geq 0$. What are the coordinates of the lowest point on the particle's trajectory (absolute minimum y-coordinate)?

A $(0, 0)$

B $(2, -16)$

C $(4, -16)$

D $(12, -16)$

ITEM 56

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A particle moves in the plane such that its speed is constant: $||v(t)|| = c$ for all t, where c is a non-zero constant. Which of the following statements MUST be true?

A The position vector is always orthogonal to the velocity vector

B The acceleration vector is the zero vector

C The particle must travel in a straight line

D The acceleration vector is always orthogonal to the velocity vector ($v(t) \cdot a(t) = 0$)

ITEM 57

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle travels along the path $r(t) = \langle 2t, t^2 \rangle$ for $t \geq 0$. At what time t is the rate of change of the angle of motion θ with respect to time, $d\theta/dt$, maximized?

A $t = 0$
B $t = 1$
C $t = 2$
D $t = \sqrt{3}$

ITEM 58

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with velocity components $x'(t) = 4t - 8$ and $y'(t) = 3$ for $t \geq 0$. At what time t is the particle's speed minimized?

A $t = 0$
B $t = 2$
C $t = 4$
D $t = 8/3$

ITEM 59

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves with position vector $r(t) = \langle \cos(\pi t), \sin(\pi t) \rangle$. What is the total distance traveled by the particle from $t = 0$ to $t = 4$?

A 8π
B 2π
C 4π
D 0

ITEM 60

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

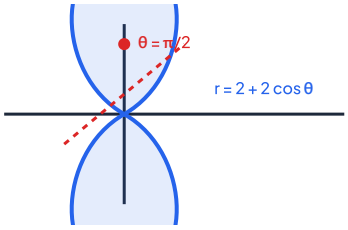
A particle moves with acceleration vector $a(t) = \langle 4, 12t \rangle$ for $t \geq 0$. If the particle is at rest at time $t = 0$ at the point $(1, -2)$, what is the particle's position vector at time $t = 1$?

A $\langle 3, -2 \rangle$
B $\langle 2, 2 \rangle$
C $\langle 5, 2 \rangle$
D $\langle 3, 0 \rangle$

ITEM 61

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the slope of the line tangent to the cardioid $r = 2 + 2\cos\theta$ at the point where $\theta = \pi / 2$?



A 1

B −1

C 0

D Undefined

ITEM 62

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the slope of the line tangent to the polar curve $r = 4\sin\theta$ at the point where $\theta = \pi / 3$?

A $\sqrt{3}$

B $-\sqrt{3}$

C $-1 / \sqrt{3}$

D $1 / \sqrt{3}$

ITEM 63

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

For the polar curve $r = 3 - 2\sin\theta$, which statement correctly interprets the value of $dr/d\theta$ at $\theta = \pi / 6$?

A $dr/d\theta = -1$, meaning the curve is moving downward

B $dr/d\theta = -\sqrt{3}$, meaning the slope of the tangent line is negative

C $dr/d\theta = -\sqrt{3}$, meaning the distance between the curve and the origin is decreasing

D $dr/d\theta = \sqrt{3}$, meaning the distance from the origin is increasing

ITEM 64

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

At which of the following values of θ in the interval $[0, 2\pi)$ does the cardioid $r = 1 + \sin\theta$ have a horizontal tangent line?

A $\theta = \pi$

B $\theta = 3\pi / 2$

C $\theta = 0$

D $\theta = \pi / 2$

ITEM 65

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A polar curve is defined by $r = 2\cos\theta$ for $0 \le \theta < \pi$. At which value of θ does the curve have a vertical tangent line?

A $\theta = 0$

B $\theta = \pi / 4$

C $\theta = 3\pi / 4$

D $\theta = \pi / 3$

ITEM 66

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

For a polar curve $r = f(\theta)$, if $f(\alpha) = 0$ and $f'(\alpha) \neq 0$, what is the line tangent to the curve at the pole (origin)?

A The line $y = \alpha$

B The line $\theta = \alpha$

C The line $x = \alpha$

D The line $y = f'(\alpha)\,x$

ITEM 67

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the Cartesian equation of the line tangent to the circle $r = 2$ at the point where $\theta = \pi / 4$?

A $x + y = 2$

B $x - y = 0$

C $x + y = 2\sqrt{2}$

D $y = x + \sqrt{2}$

ITEM 68

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

For the polar curve $r = \theta^2$, what is the rate of change of y with respect to θ ($dy / d\theta$) at $\theta = \pi$?

A π^2

B 2π

C 0

D $-\pi^2$

ITEM 69

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Values of a polar curve r(θ) and its derivative are given in the table below.

θ	r(θ)	dr/dθ
π/4	4	2
π/2	3	−1
3π/4	1	−4

What is the slope of the line tangent to the curve at $\theta = \pi / 4$?

A −3

B 3

C $-1/3$

D 1

ITEM 70

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

At what value(s) of θ in the interval $[0, 2\pi)$ does the limaçon $r = 1 - 2\cos\theta$ pass through the pole (origin)?

A $\theta = \pi/6$ and $\theta = 11\pi/6$

B $\theta = \pi/3$ and $\theta = 5\pi/3$

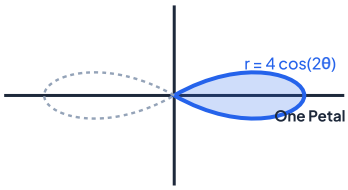
C $\theta = 2\pi/3$ and $\theta = 4\pi/3$

D $\theta = 0$

ITEM 71

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area enclosed by one petal of the four-petaled rose $r = 4\cos(2\theta)$?



A π

B 4π

C 2π

D 8π

ITEM 72

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the total area enclosed by the circle $r = 6\sin\theta$?

A $9\pi/2$

B 18π

C 36π

D 9π

ITEM 73

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the total area enclosed by the cardioid $r = 3 + 3\sin\theta$?

A $27\pi / 2$
B 27π
C 9π
D 18π

ITEM 74

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area enclosed by one petal of the three-petaled rose $r = 2\sin(3\theta)$?

A $2\pi / 3$
B $\pi / 3$
C $\pi / 6$
D π

ITEM 75

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the inner loop of the limaçon $r = 1 - 2\sin\theta$?

A $\pi / 2 - (3\sqrt{3} / 4)$
B $2\pi - 3\sqrt{3}$
C $\pi - (3\sqrt{3} / 2)$
D $\pi + (3\sqrt{3} / 2)$

ITEM 76

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

Which of the following definite integrals represents the area of the region bounded by the logarithmic spiral $r = e^{2\theta}$ for $0 \le \theta \le \pi / 2$?

A $\int_0^{\pi/2} e^{2\theta}\, d\theta$
B $\int_0^{\pi/2} e^{4\theta}\, d\theta$
C $(1/2)\int_0^{\pi/2} e^{2\theta}\, d\theta$
D $(1/2)\int_0^{\pi/2} e^{4\theta}\, d\theta$

ITEM 77

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the total area enclosed by both loops of the lemniscate $r^2 = 9\cos(2\theta)$?

A 9
B 18
C 9 / 2
D 9π

ITEM 78

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A ray $\theta = k$ (where $0 < k < \pi$) divides the area enclosed by the cardioid $r = 2 + 2\cos\theta$ in the upper half-plane ($0 \le \theta \le \pi$) into two equal parts. Which equation can be used to solve for k?

A $\int_0^k (2 + 2\cos\theta)^2\, d\theta = 6\pi$
B $\int_0^k (2 + 2\cos\theta)^2\, d\theta = 3\pi$
C $\int_0^k (2 + 2\cos\theta)\, d\theta = 3\pi$
D $\int_0^k (2 + 2\cos\theta)^2\, d\theta = 3\pi / 2$

ITEM 79

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The radius $r(\theta)$ of a polar curve is given in the table below at selected angles.

θ	0	π/6	π/3	π/2
r(θ)	2	√7	√7	2

Using a trapezoidal sum with the three subintervals indicated by the table, what is the approximation for the area bounded by the curve from $\theta = 0$ to $\theta = \pi / 2$?

A 3π / 4
B 3π
C 3π / 2
D 5π / 3

ITEM 80

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

What is the area of the circular sector bounded by $r = 4$ and the rays $\theta = \pi / 6$ and $\theta = 2\pi / 3$?

A 16π
B 2π
C 8π
D 4π

ITEM 81

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Which of the following definite integrals gives the area of the region that lies inside the circle $r = 3$ and outside the cardioid $r = 2 + 2\cos\theta$?

A $(1/2)\int_{\pi/3}^{5\pi/3}[9 - (2+2\cos\theta)^2]\,d\theta$

B $(1/2)\int_{-\pi/3}^{3\pi/3}[9 - (2+2\cos\theta)^2]\,d\theta$

C $\int_{\pi/3}^{5\pi/3}[3 - (2+2\cos\theta)]\,d\theta$

D $(1/2)\int_{\pi/3}^{5\pi/3}[(2+2\cos\theta)^2 - 9]\,d\theta$

ITEM 82

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region common to both circles $r = 2\sin\theta$ and $r = 2\cos\theta$ in the first quadrant $0 \le \theta \le \pi/2$?

A $\pi - 2$

B $\pi/2 - 1$

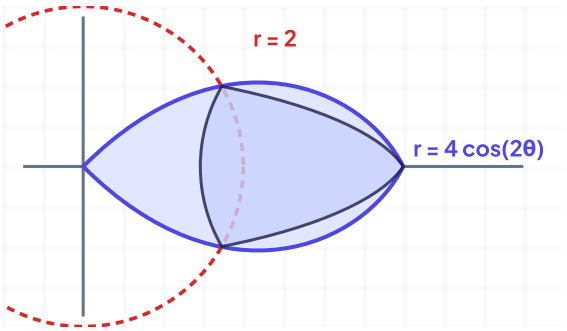
C $\pi/4$

D $\pi/2$

ITEM 83

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The figure below shows the region inside the right petal of the four-leaf rose $r = 4\cos(2\theta)$ and outside the circle $r = 2$.



What is the area of this shaded region?

ITEM 84

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the exact length of the polar curve $r = \theta^2$ for $0 \le \theta \le \sqrt{5}$?

A 19

B 38/3

C 8/3

D 19/3

ITEM 85

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the total arc length of the logarithmic spiral $r = e^{3\theta}$ from $\theta = 0$ to $\theta = 2$?

A $(\sqrt{10}/3)(e^6 - 1)$

B $\sqrt{10}(e^6 - 1)$

C $(1/3)(e^6 - 1)$

D $(\sqrt{10}/6)(e^6 - 1)$

ITEM 86

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle travels along the polar curve $r = 4\cos\theta$ in such a way that $d\theta/dt = 3$ radians per second. At the instant when $\theta = \pi/3$, at what rate is the area swept out by the position vector with respect to time t?

A 12

B 6

C 3

D 18

ITEM 87

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the total perimeter of the closed region bounded by the upper half of the cardioid $r = 1 + \cos\theta$ (for $0 \le \theta \le \pi$) and the line segment along the x-axis connecting the pole to the curve's x-intercept?

A $2 + \pi$

B 4

C 6

D $4 + \pi$

ITEM 88

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A smooth polar curve $r = f(\theta)$ has selected values of $f(\theta)$ and $f'(\theta)$ given in the table below.

θ	0	$\pi/4$	$\pi/2$
$f(\theta)$	3	6	5
$f'(\theta)$	4	8	12

Using a trapezoidal sum with the two subintervals indicated by the table, what is the approximation for the length of the curve from $\theta = 0$ to $\theta = \pi/2$?

A $19\pi/2$

B $19\pi/8$

C 38π

D $19\pi/4$

ITEM 89

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region enclosed by the inner loop of the limaçon $r = 1 + 2\cos\theta$?

A $\pi - (3\sqrt{3}/2)$

B $2\pi - 3\sqrt{3}$

C $\pi/2 - \sqrt{3}/4$

D $\pi + (3\sqrt{3}/2)$

ITEM 90

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region that lies inside the circle $r = 4\sin\theta$ and outside the circle $r = 2$?

A $2\pi/3 + \sqrt{3}$

B $4\pi/3 + 2\sqrt{3}$

C $4\pi/3 - 2\sqrt{3}$

D $8\pi/3$

FRQ 1: Parametric Motion & Velocity Table Analysis

★★★★★ VERY IMPORTANT | 9 Points

A particle moves along a smooth curve in the xy-plane for $0 \le t \le 12$. Its position at time t is given by $(x(t), y(t))$, where x and y are twice-differentiable functions of t. Selected values of $x(t)$, $x'(t)$, $y(t)$, and $y'(t)$ are shown in the table below.

t	0	3	6	9	12
x(t)	1	4	9	12	8
x'(t)	2	5	0	−3	−4
y(t)	5	3	8	14	10
y'(t)	−1	2	4	0	−2

(a) [2 Points] Find the slope of the tangent line to the path of the particle at $t = 3$. Write an equation for the line tangent to the path at the point where $t = 3$.

(b) [2 Points] Find the speed of the particle at $t = 0$. Given that $x''(0) = 3$ and $y''(0) = 2$, determine whether the speed of the particle is increasing or decreasing at $t = 0$. Justify your answer.

(c) [2 Points] At $t = 6$, does the path of the particle have a horizontal tangent line, a vertical tangent line, or neither? Justify your answer.

(d) [3 Points] Use a trapezoidal sum with the four subintervals indicated by the table to approximate the total distance traveled by the particle over the time interval $0 \le t \le 12$.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 2: Planar Vector Motion & Particle Kinematics

★★★★★ VERY IMPORTANT 9 Points

A particle moves in the xy-plane so that its velocity vector for $t \geq 0$ is given by $v(t) = \langle 3t^2 - 12, \ 4e^{2t-4} - 4 \rangle$. At time $t = 2$, the particle is located at the position $(x(2), y(2)) = (5, -3)$.

(a) [3 Points] Find the position vector $r(t) = \langle x(t), y(t) \rangle$ of the particle at any time $t \geq 0$.

(b) [2 Points] Find all times $t > 0$ at which the particle is at rest.

(c) [2 Points] Find the acceleration vector $a(2)$ at $t = 2$. Find the direction angle θ $(0 \leq \theta < 2\pi)$ of the acceleration vector at this instant.

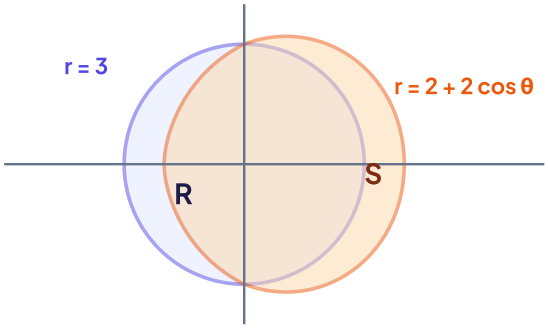
(d) [2 Points] Write, but do not evaluate, an expression involving one or more integrals that gives the total distance traveled by the particle from $t = 1$ to $t = 3$.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 3: Polar Area Bounded by Two Curves

★★★★★ VERY IMPORTANT 9 Points

The figure below shows the graphs of the circle $r = 3$ and the cardioid $r = 2 + 2\cos\theta$ in polar coordinates for $0 \le \theta \le 2\pi$. Region R lies inside the circle and outside the cardioid. Region S lies inside the cardioid and outside the circle.



(a) [2 Points] Find the polar coordinates (r, θ) of all intersection points of the two curves on the interval $0 \le \theta < 2\pi$.

(b) [3 Points] Find the exact area of region R.

(c) [2 Points] Write, but do not evaluate, an integral expression that gives the area of region S.

(d) [2 Points] A ray $\theta = k$ $(0 < k < 2\pi)$ divides the entire area enclosed by the cardioid into two regions of equal area. Find the value of k and justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 4: Polar Slopes & Tangent Lines for a Limacon

★★★★★ VERY IMPORTANT 9 Points

Consider the polar curve defined by $r = 2 - 2\sin\theta$ for $0 \le \theta \le 2\pi$.

(a) [3 Points] Show that $dy/d\theta = 2\cos\theta(1 - 2\sin\theta)$ and $dx/d\theta = -2\sin\theta - 2\cos(2\theta)$. Use these derivatives to find the slope of the tangent line to the curve at $\theta = \pi / 6$.

(b) [2 Points] Find the Cartesian coordinates (x, y) of the point on the curve corresponding to $\theta = \pi / 6$, and write an equation for the tangent line to the curve at this point.

(c) [2 Points] Find all values of θ in $0 \le \theta < 2\pi$ at which the tangent line to the curve is horizontal.

(d) [2 Points] Find the angle θ where the curve passes through the pole (origin), and determine the equation of the tangent line to the curve at the pole.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 5: Applied Drone Flight & Parametric Arc Length

★★★★★ VERY IMPORTANT 9 Points

An autonomous surveillance drone flies in a flat horizontal proving ground. For $0 \le t \le 10$ seconds, the drone's position is given by the parametric equations $(x(t), y(t))$, with velocity components $x'(t) = 6\sqrt{t}$ and $y'(t) = 3t - 3$. At time $t = 0$, the drone is located at $(x(0), y(0)) = (0, 4)$.

(a) [2 Points] Find the position of the drone at $t = 4$ seconds.

(b) [2 Points] Find the speed of the drone at $t = 4$ seconds.

(c) [3 Points] Find the exact total distance traveled by the drone from $t = 0$ to $t = 4$ seconds.

(d) [2 Points] Write, but do not evaluate, an integral expression that gives the total distance traveled by the drone from $t = 4$ to $t = 10$ seconds.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 6: Vector Acceleration & Maximum Height Flight

★★★★★ VERY IMPORTANT 9 Points

A projectile moves in the coordinate plane. The acceleration vector of the projectile for $t \geq 0$ is given by $a(t) = \langle -2, \ 6 - 6t \rangle$. At time $t = 0$, the projectile is launched from the origin $(0, 0)$ with initial velocity $v(0) = \langle 10, 9 \rangle$.

(a) [3 Points] Find the velocity vector $v(t)$ and position vector $r(t)$ of the projectile at any time $t \geq 0$.

(b) [2 Points] Find the time $t > 0$ at which the projectile reaches its maximum vertical height, and find this maximum height.

(c) [2 Points] Find the speed and the acceleration magnitude of the projectile at the instant of maximum height.

(d) [2 Points] At the instant of maximum height, is the speed of the projectile increasing, decreasing, or neither? Justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 7: Polar Rose Curve Petals, Perimeter & Radial Rates

★★★★★ VERY IMPORTANT

9 Points

Consider the four-leaf rose curve defined by $r = 3\sin(2\theta)$ for $0 \le \theta \le 2\pi$.

(a) [3 Points] Find the interval of θ for which the petal in Quadrant I is traced, and find the area enclosed by this single petal.

(b) [2 Points] What is the total area enclosed by all four petals of the rose curve combined?

(c) [2 Points] Write, but do not evaluate, an integral expression that gives the perimeter (arc length) of the petal in Quadrant I.

(d) [2 Points] At $\theta = \pi / 6$, find $dr/d\theta$ and determine whether the distance from the point on the curve to the pole is increasing or decreasing. Justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 8: Two Particles — Collision vs. Intersecting Paths

★★★★★ VERY IMPORTANT 9 Points

Two particles, Particle A and Particle B, move in the xy-plane for time $t \geq 0$. Their positions at time t are given by:
Particle A: $x_A(t) = 2t - 1$, $y_A(t) = t^2 - 4t + 7$
Particle B: $x_B(t) = t^2 - 4$, $y_B(t) = 3t - 2$

(a) [3 Points] Do the particles ever collide? If so, find the time t and position of collision. If not, explain why.

(b) [3 Points] Find the Cartesian (rectangular) equation of the path traveled by Particle A in terms of x and y.

(c) [3 Points] At time $t = 3$, which particle is moving faster? Justify your answer by calculating the speed of each particle.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 9: Applied Vector Rate Modeling — Oceanic Vessel Navigation

★★★★★ VERY IMPORTANT

9 Points

A research vessel navigates in coastal waters for $0 \le t \le 12$ hours. The vessel's engine generates a velocity relative to the water given by $v_V(t) = \langle 4, \ 3\cos(t/2) \rangle$ knots. Meanwhile, an ocean tidal current exerts a velocity on the vessel of $v_C(t) = \langle 2\sin(t/4), \ -1 \rangle$ knots. The net velocity of the vessel is $v(t) = v_V(t) + v_C(t) = \langle 4 + 2\sin(t/4), \ 3\cos(t/2) - 1 \rangle$. At time $t = 0$, the vessel is located at the origin $(0, 0)$.

(a) [2 Points] Find the net velocity vector and the speed of the vessel at time $t = 0$.

(b) [3 Points] Find the position $(x(2\pi), y(2\pi))$ of the vessel at time $t = 2\pi$ hours.

(c) [2 Points] Find the acceleration vector $a(2\pi)$ of the vessel at time $t = 2\pi$ hours.

(d) [2 Points] Write, but do not evaluate, an integral expression that gives the total distance traveled by the vessel over the entire 12-hour period.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 10: Comprehensive Unit 9 Capstone Synthesis

★★★★★ VERY IMPORTANT 9 Points

A curve C is defined in polar coordinates by $r(\theta) = 2 + 4\cos\theta$ for $0 \le \theta \le 2\pi$. A particle moves along the curve C in such a way that the angle θ increases at a constant rate of $d\theta/dt = 2$ radians per second.

(a) [2 Points] At the instant when $\theta = \pi / 3$, find the Cartesian coordinates (x, y) of the particle and find the rate of change dr/dt of the radius with respect to time.

(b) [3 Points] At the instant when $\theta = \pi / 3$, find the velocity vector $v = \langle dx/dt, dy/dt \rangle$ of the particle and calculate the speed of the particle.

(c) [2 Points] Find the slope of the line tangent to the path of the particle at $\theta = \pi / 3$.

(d) [2 Points] Write, but do not evaluate, an integral expression that gives the total area enclosed by the inner loop of the limaçon.

Official AP Scoring Directive: Show all supporting work. Clearly identify parametric/polar derivatives, integral limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

End of Question Paper

You have concluded Questions 1 through 100 of the Unit 9 Master Examination Repository.

> **Important Study Strategy:** Carefully compare your solutions against the detailed answer keys and explanations that follow. Pay special attention to why incorrect choices were selected as distractors and how full credit is awarded on official AP rubrics.

Answer Section Contents

- **Answer Key Table:** Complete quick-reference grid of letter choices for Questions 1–90.
- **MCQ Detailed Solutions:** Step-by-step bulleted solutions for Questions 1–90 with full distractor error analysis.
- **FRQ Model Solutions:** Full analytical solutions for Questions 91–100 with official AP 9-point rubric point breakdowns.
- **Formula Appendix:** Summary of high-yield Unit 9 parametric equations, polar coordinates, and vector motion theorems.

Multiple Choice Answer Key

Verified answer keys with priority ratings. Questions 91–100 are free response (90 points total).

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
A ★★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★	B ★★★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
C ★★★★★	D ★★★★★	A ★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
A ★★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
C ★★★★★	D ★★★★	A ★★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★★	C ★★★	D ★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
A ★★★★★	B ★★★★	C ★★★	D ★★★★	A ★★★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★	B ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
C ★★★★★	D ★★★★★	A ★★★★★	B ★★★★	C ★★★★	D ★★★	A ★★★★★	B ★★★★	C ★★★★	D ★★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★	C ★★★★	D ★★★	A ★★★★	B ★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★	A ★★★★	B ★★★★★	C ★★★★	D ★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
A ★★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★★

Scoring Summary: Each multiple choice question is worth 1 point (90 points total). Each free response question is worth 9 points (90 points total). Total Exam Value: 180 Points.

Question 01 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 4 / 9

STEP-BY-STEP SOLUTION

- **Recall the Parametric Derivative Formula:** $dy/dx = (dy/dt) / (dx/dt)$.
- **Differentiate x(t) with respect to t:**
- • $x(t) = t^3 - 3t \Rightarrow dx/dt = 3t^2 - 3$.
- • At $t = 2$: $dx/dt = 3(2)^2 - 3 = 3(4) - 3 = 12 - 3 = 9$.
- **Differentiate y(t) with respect to t:**
- • $y(t) = t^2 + 2 \Rightarrow dy/dt = 2t$.
- • At $t = 2$: $dy/dt = 2(2) = 4$.
- **Compute the Tangent Slope dy/dx:**
- • $dy/dx = (dy/dt) / (dx/dt) = 4 / 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates the reciprocal $(dx/dt) / (dy/dt) = 9 / 4$ instead of $(dy/dt) / (dx/dt)$.

✗ **(C)** Omits the -3 term when differentiating x(t), calculating $4 / 3(4) = 4 / 12 = 1/3$ or arithmetic mistake.

✗ **(D)** Differentiates y(t) as 2 instead of $2t$, arriving at $2 / 3$.

Question 02 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) y = (3/2)(x – 1)

STEP-BY-STEP SOLUTION

- **Find the Coordinates (x, y) at t = 0:**
- • $x(0) = e^{2 \cdot 0} = e^0 = 1$.
- • $y(0) = \sin(3 \cdot 0) = \sin(0) = 0$.
- • Tangent point is $(x_0, y_0) = (1, 0)$.
- **Compute Derivatives with respect to t:**
- • $dx/dt = 2e^{2t} \Rightarrow$ at $t = 0$: $dx/dt = 2e^0 = 2$.
- • $dy/dt = 3\cos(3t) \Rightarrow$ at $t = 0$: $dy/dt = 3\cos(0) = 3(1) = 3$.
- **Determine Slope dy/dx:**
- • $m = dy/dx = (dy/dt) / (dx/dt) = 3 / 2$.
- **Write Point-Slope Tangent Equation:**
- • $y - y_0 = m(x - x_0) \Rightarrow y - 0 = (3/2)(x - 1) \Rightarrow y = (3/2)(x - 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the slope: $m = (dx/dt) / (dy/dt) = 2/3$.

✗ **(C)** Forgets the chain rule coefficient 2 on dx/dt, using slope $3 / 1 = 3$.

Question 03 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) t = 0 only

STEP-BY-STEP SOLUTION

- **Condition for a Horizontal Tangent:**
- • A parametric curve has a horizontal tangent where $dy/dt = 0$ and $dx/dt \neq 0$.
- **Compute Derivatives:**
- • $dy/dt = 4t$.
- • $dx/dt = 3t^2 - 12 = 3(t^2 - 4) = 3(t - 2)(t + 2)$.
- **Find Where dy/dt = 0:**
- • $4t = 0 \Rightarrow t = 0$.
- **Verify dx/dt at t = 0:**
- • $dx/dt = 3(0)^2 - 12 = -12 \neq 0$.
- • Since $dy/dt = 0$ and $dx/dt \neq 0$ at $t = 0$, the curve has a horizontal tangent at $t = 0$ only.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes points where $dx/dt = 0$ as well as $dy/dt = 0$ without checking which derivative must vanish.

✗ **(B)** Confuses horizontal tangents with vertical tangents (where $dx/dt = 0 \Rightarrow t = \pm 2$).

✗ **(D)** Believes there are no horizontal tangents because the curve is non-linear.

Question 04 Analysis

★★★★ MANDATORY

Correct Answer: (D) t = 2 only

STEP-BY-STEP SOLUTION

- **Condition for a Vertical Tangent:**
- • A parametric curve has a vertical tangent where dx/dt = 0 and dy/dt ≠ 0.
- **Compute Derivatives:**
- • dx/dt = 2t – 4.
- • dy/dt = $3t^2$ – 6.
- **Solve dx/dt = 0:**
- • 2t – 4 = 0 ⟹ 2t = 4 ⟹ t = 2.
- **Check dy/dt at t = 2:**
- • dy/dt = $3(2)^2$ – 6 = 12 – 6 = 6 ≠ 0.
- • Since dx/dt = 0 and dy/dt = 6 ≠ 0 at t = 2, the curve has a vertical tangent line at t = 2 only.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Combines points where dx/dt = 0 and dy/dt = 0 without distinguishing vertical from horizontal tangents.
✗ **(B)** Solves dy/dt = 0 ⟹ $3t^2$ – 6 = 0 ⟹ t = ±√2 (which gives horizontal tangents).
✗ **(C)** Solves x(t) = 0 ⟹ t(t – 4) = 0, confusing x-intercepts with vertical tangents.

Question 05 Analysis

★★★★ MANDATORY

Correct Answer: (A) –3 / 4

STEP-BY-STEP SOLUTION

- **Compute Component Derivatives:**
- • dx/dt = –4 sin t.
- • dy/dt = 3 cos t.
- **Compute Tangent Slope dy/dx:**
- • dy/dx = (dy/dt) / (dx/dt) = (3 cos t) / (–4 sin t) = –(3/4) cot t.
- **Evaluate at t = π / 4:**
- • cot(π / 4) = 1.
- • dy/dx = –(3/4)(1) = –3 / 4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in derivative of cosine: forgets the negative sign, yielding +3/4.
✗ **(C)** Inverts the fraction: –4/3.
✗ **(D)** Inverts the fraction and drops the negative sign: 4/3.

Question 06 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 3

STEP-BY-STEP SOLUTION

- **Identify Target Slope:**
- • 2x – y = 5 ⟹ y = 2x – 5, so the required slope is m = 2.
- **Find Derivatives of x and y with respect to t:**
- • dx/dt = 1 – 1/t = (t – 1) / t.
- • dy/dt = 1 + 1/t = (t + 1) / t.
- **Compute dy/dx:**
- • dy/dx = (dy/dt) / (dx/dt) = [(t + 1) / t] / [(t – 1) / t] = (t + 1) / (t – 1) for t ≠ 1.
- **Set dy/dx = 2 and Solve for t:**
- • (t + 1) / (t – 1) = 2 ⟹ t + 1 = 2(t – 1) = 2t – 2.
- • t + 1 = 2t – 2 ⟹ 2t – t = 1 + 2 ⟹ t = 3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculation error: t + 1 = 2t – 1 ⟹ t = 2.
✗ **(C)** Solves dx/dt = 0 ⟹ t = 1, where the derivative is undefined.
✗ **(D)** Reciprocal error: solves (t – 1) / (t + 1) = 2.

Question 07 Analysis

★★★★ MANDATORY

Correct Answer: (C) $y - 1 = -2(x - 5)$

STEP-BY-STEP SOLUTION

- **Identify Coordinates at t = 2:**
- • From the table, $x(2) = 5$ and $y(2) = 1$.
- • Point of tangency is $(5, 1)$.
- **Compute Slope dy/dx at t = 2:**
- • From the table, $x'(2) = -3$ and $y'(2) = 6$.
- • $dy/dx = y'(2) / x'(2) = 6 / (-3) = -2$.
- **Write Tangent Line Equation:**
- • $y - y_1 = m(x - x_1) \Rightarrow y - 1 = -2(x - 5)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the slope: $-3 / 6 = -1/2$.
✗ **(B)** Sign error in slope: calculates $6 / 3 = 2$ instead of $6 / (-3) = -2$.
✗ **(D)** Uses the point at $t = 1$ instead of $t = 2$: $(3, -2)$.

Question 08 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (2, –2) and (2, 2)

STEP-BY-STEP SOLUTION

- **Horizontal Tangent Condition (Slope = 0):** $dy/dx = 0 \Rightarrow dy/dt = 0$ with $dx/dt \neq 0$.
- **Compute Derivatives:**
- • $dx/dt = 2t$.
- • $dy/dt = 3t^2 - 3 = 3(t^2 - 1) = 3(t - 1)(t + 1)$.
- **Solve dy/dt = 0:**
- • $3(t - 1)(t + 1) = 0 \Rightarrow t = 1$ or $t = -1$.
- **Verify dx/dt ≠ 0:**
- • At $t = 1$: $dx/dt = 2(1) = 2 \neq 0$.
- • At $t = -1$: $dx/dt = 2(-1) = -2 \neq 0$.
- **Find Corresponding Points (x, y):**
- • When $t = 1$: $x = (1)^2 + 1 = 2$, $y = (1)^3 - 3(1) = 1 - 3 = -2 \Rightarrow (2, -2)$.
- • When $t = -1$: $x = (-1)^2 + 1 = 2$, $y = (-1)^3 - 3(-1) = -1 + 3 = 2 \Rightarrow (2, 2)$.
- • The points are $(2, -2)$ and $(2, 2)$.

Question 09 Analysis

★★★ IMPORTANT

Correct Answer: (A) csc θ

STEP-BY-STEP SOLUTION

- **Differentiate x and y with respect to θ:**
- • $dx/d\theta = 5 \sec\theta \tan\theta$.
- • $dy/d\theta = 5 \sec^2\theta$.
- **Compute dy/dx:**
- • $dy/dx = (dy/d\theta) / (dx/d\theta) = (5 \sec^2\theta) / (5 \sec\theta \tan\theta) = \sec\theta / \tan\theta$.
- **Simplify Trigonometric Quotient:**
- • $\sec\theta = 1 / \cos\theta$ and $\tan\theta = \sin\theta / \cos\theta$.
- • $\sec\theta / \tan\theta = (1 / \cos\theta) \cdot (\cos\theta / \sin\theta) = 1 / \sin\theta = \csc\theta$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Inverts trigonometric quotient, writing $(\sin\theta / \cos\theta) / (1 / \cos\theta) = \sin\theta$.
✗ **(C)** Confuses $\csc\theta$ with $\cos\theta$.
✗ **(D)** Cancels $\tan\theta$ incorrectly, leaving $\sec\theta$.

Question 10 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1

STEP-BY-STEP SOLUTION

- **Compute Derivatives:**
- • $dx/dt = 3t^2 - 6t = 3t(t - 2)$.
- • $dy/dt = 3t^2 - 3 = 3(t^2 - 1)$.
- **Set dy/dx = 1:**
- • $(dy/dt) / (dx/dt) = 1 \Rightarrow dy/dt = dx/dt$ (with $dx/dt \neq 0$).
- • $3t^2 - 3 = 3t^2 - 6t$.
- • Subtract $3t^2$ from both sides: $-3 = -6t \Rightarrow 6t = 3 \Rightarrow t = 1/2$.
- **Verify dx/dt at t = 1/2:**
- • $dx/dt = 3(1/2)(1/2 - 2) = (3/2)(-3/2) = -9/4 \neq 0$.
- • Since there is exactly one value of $t = 1/2$ where $dy/dx = 1$, there is exactly 1 such point on the curve.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Incorrectly treats quadratic equation as having two solutions.

✗ **(C)** Counts points where $dy/dt = 0$ or $dx/dt = 0$.

✗ **(D)** Assumes $dy/dx = 1$ has no solutions because $3t^2$ cancels.

Question 11 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 3 / 2

STEP-BY-STEP SOLUTION

- **Recall the Formula for the Second Parametric Derivative:**
- • $d^2y/dx^2 = [\, d/dt\,(dy/dx)\,] / (dx/dt)$.
- **Step 1: Compute First Derivative dy/dx:**
- • $dx/dt = 2t$.
- • $dy/dt = 3t^2 - 3$.
- • $dy/dx = (3t^2 - 3) / (2t) = (3/2)\,t - (3/2)\,t^{-1}$.
- **Step 2: Differentiate dy/dx with respect to t:**
- • $d/dt\,(dy/dx) = 3/2 - (3/2)(-1)\,t^{-2} = 3/2 + 3 / (2t^2)$.
- • At $t = 1$: $d/dt\,(dy/dx) = 3/2 + 3/2 = 3$.
- **Step 3: Divide by dx/dt:**
- • At $t = 1$: $dx/dt = 2(1) = 2$.
- • $d^2y/dx^2 = 3 / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by $(dx/dt)^2$ instead of dx/dt: 3 / 4

Question 12 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) –8

STEP-BY-STEP SOLUTION

- **Compute First Derivatives:**
- • $dx/dt = -\sin t$.
- • $dy/dt = \cos t$.
- • $dy/dx = (dy/dt) / (dx/dt) = (\cos t) / (-\sin t) = -\cot t$.
- **Differentiate dy/dx with respect to t:**
- • $d/dt\,(-\cot t) = -(-\csc^2 t) = \csc^2 t$.
- **Apply Second Derivative Formula:**
- • $d^2y/dx^2 = [\, d/dt\,(dy/dx)\,] / (dx/dt) = (\csc^2 t) / (-\sin t) = -\csc^3 t$.
- **Evaluate at $t = \pi / 6$:**
- • $\sin(\pi / 6) = 1/2 \Rightarrow \csc(\pi / 6) = 2$.
- • $d^2y/dx^2 = -(2)^3 = -8$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses d^2y/dx^2 with curvature or assumes circle has constant derivative –1.

✗ **(B)** Forgets to divide by dx/dt, calculating only $d/dt\,(dy/dx) = \csc^2(\pi/6) = 4$, with sign error giving –4.

✗ **(C)** Sign error dropping the negative sign: +8.

Question 13 Analysis

★★★★ MANDATORY

Correct Answer: (A) 2 / 9

STEP-BY-STEP SOLUTION

- **Find dx/dt and dy/dt:**
- • $dx/dt = 3t^2$.
- • $dy/dt = 2t - 2$.
- **Compute First Derivative dy/dx:**
- • $dy/dx = (2t - 2) / (3t^2) = (2/3)\,t^{-1} - (2/3)\,t^{-2}$.
- **Differentiate dy/dx with respect to t:**
- • $d/dt\,(dy/dx) = -(2/3)\,t^{-2} + (4/3)\,t^{-3} = (-2t + 4) / (3t^3)$.
- • At t = 1: $d/dt\,(dy/dx) = (-2(1) + 4) / 3(1)^3 = 2 / 3$.
- **Apply Parametric Second Derivative Formula:**
- • At t = 1: $dx/dt = 3(1)^2 = 3$.
- • $d^2y/dx^2 = [\,d/dt\,(dy/dx)\,] / (dx/dt) = (2 / 3) / 3 = 2 / 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in differentiating negative powers: –2/9.

✗ **(C)** Forgets to divide by dx/dt = 3 in the final step: reports 2/3.

Question 14 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) t > 1

STEP-BY-STEP SOLUTION

- **Compute Component Derivatives:**
- • $dx/dt = e^t$.
- • $dy/dt = 2t$.
- **Find dy/dx:**
- • $dy/dx = (2t) / e^t = 2t\,e^{-t}$.
- **Differentiate with respect to t using Product Rule:**
- • $d/dt\,[\,2t\,e^{-t}\,] = 2\,e^{-t} - 2t\,e^{-t} = 2(1 - t)\,e^{-t}$.
- **Compute Second Derivative:**
- • $d^2y/dx^2 = [\,2(1 - t)\,e^{-t}\,] / e^t = 2(1 - t)\,e^{-2t}$.
- **Determine Sign of d^2y/dx^2:**
- • Since $e^{-2t} > 0$ for all t, the sign depends strictly on (1 – t).
- • Concave downward requires $d^2y/dx^2 < 0 \Rightarrow 1 - t < 0 \Rightarrow t > 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Identifies the interval where the curve is concave upward (t < 1).

Question 15 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) (6, 2)

STEP-BY-STEP SOLUTION

- **Condition for Point of Inflection:**
- • An inflection point occurs where $d^2y/dx^2 = 0$ and changes sign.
- **Compute Derivatives:**
- • $dx/dt = 3$.
- • $dy/dt = 3t^2 - 12t + 9$.
- • $dy/dx = (3t^2 - 12t + 9) / 3 = t^2 - 4t + 3$.
- **Find Second Derivative:**
- • $d/dt\,(dy/dx) = 2t - 4$.
- • $d^2y/dx^2 = (2t - 4) / (dx/dt) = (2t - 4) / 3$.
- **Solve $d^2y/dx^2 = 0$:**
- • $(2t - 4) / 3 = 0 \Rightarrow 2t = 4 \Rightarrow t = 2$.
- • Since d^2y/dx^2 changes from negative to positive at t = 2, it is a point of inflection.
- **Find Coordinates (x, y) at t = 2:**
- • $x(2) = 3(2) = 6$.
- • $y(2) = (2)^3 - 6(2)^2 + 9(2) = 8 - 24 + 18 = 2$.
- • Inflection point is (6, 2).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates at t = 1 (where dy/dx = 0): (3, 4).

✗ **(B)** Uses t = 2 directly as the x-coordinate: (2, 2).

✗ **(D)** Evaluates y(2) with arithmetic error yielding 0: (6, 0).

Question 16 Analysis

★★★★ MANDATORY

Correct Answer: (D) 2

STEP-BY-STEP SOLUTION

- **Compute Correct First Derivative:**
- • $dx/dt = 2t$.
- • $dy/dt = 4t^3$.
- • $dy/dx = (4t^3) / (2t) = 2t^2$.
- **Compute Correct Second Derivative:**
- • $d/dt\,(dy/dx) = d/dt\,[2t^2] = 4t$.
- • $d^2y/dx^2 = [\,d/dt\,(dy/dx)\,] / (dx/dt) = (4t) / (2t) = 2$ (a constant!).
- • Thus at $t = 2$, $d^2y/dx^2 = 2$.
- **Note on the Common Student Trap:**
- • The student's incorrect formula gives $(d^2y/dt^2) / (d^2x/dt^2) = (12t^2) / 2 = 6t^2$.
- • At $t = 2$, that incorrect formula yields $6(4) = 24$ (Option B).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates $4t$ at $t = 2$ without dividing by $dx/dt = 2t = 4$: 8.

✗ **(B)** Falls into the classic trap of computing $(d^2y/dt^2) / (d^2x/dt^2) = 12(4) / 2 = 24$.

Question 17 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 7 / 4

STEP-BY-STEP SOLUTION

- **Use Quotient Rule on dy/dx = y'(t) / x'(t):**
- • $d/dt\,(dy/dx) = [\,x'(t)\,y''(t) - y'(t)\,x''(t)\,] / [x'(t)]^2$.
- **Substitute Values at t = 3:**
- • $x'(3) = 2$, $y'(3) = 6$, $x''(3) = -1$, $y''(3) = 4$.
- • $d/dt\,(dy/dx) = [\,(2)(4) - (6)(-1)\,] / (2)^2 = [\,8 + 6\,] / 4 = 14 / 4 = 7 / 2$.
- **Divide by x'(3) to Obtain d^2y/dx^2:**
- • $d^2y/dx^2 = [\,d/dt\,(dy/dx)\,] / x'(3) = (7 / 2) / 2 = 7 / 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Computes $y''(3) / x'(3) = 4 / 2 = 2$, ignoring $x''(3)$ and the quotient rule.

✗ **(C)** Forgets the final division by $x'(3)$, reporting $d/dt(dy/dx) = 7/2$.

✗ **(D)** Evaluates only the numerator: 14.

Question 18 Analysis

★★★★ MANDATORY

Correct Answer: (B) 1 / 2

STEP-BY-STEP SOLUTION

- **Compute First Derivatives:**
- • $dx/dt = 1 / t$.
- • $dy/dt = (1/2)\,t^{-1/2} = 1 / (2\sqrt{t})$.
- **Find dy/dx:**
- • $dy/dx = [\,1 / (2\sqrt{t})\,] / [\,1 / t\,] = t / (2\sqrt{t}) = (1/2)\,t\text{^}(1/2) = (1/2)\,\sqrt{t}$.
- **Differentiate dy/dx with respect to t:**
- • $d/dt\,(dy/dx) = (1/2) \cdot (1/2)\,t^{-1/2} = 1 / (4\sqrt{t})$.
- **Apply Parametric Second Derivative Formula:**
- • $d^2y/dx^2 = [\,d/dt\,(dy/dx)\,] / (dx/dt) = [\,1 / (4\sqrt{t})\,] / [\,1 / t\,] = t / (4\sqrt{t}) = (1/4)\,\sqrt{t}$.
- **Evaluate at t = 4:**
- • $d^2y/dx^2 = (1/4)\,\sqrt{4} = (1/4)(2) = 1 / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects to multiply by $\sqrt{4} = 2$: 1/4.

✗ **(C)** Calculates $d/dt(dy/dx)$ at $t = 4$ without dividing by dx/dt: $1 / (4 \cdot 2) = 1/8$.

✗ **(D)** Arithmetic mistake in fraction division: 1.

Question 19 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) –1

STEP-BY-STEP SOLUTION

- **Compute Derivatives:**
- • $dx/dt = 1 - \cos t$.
- • $dy/dt = \sin t$.
- • $dy/dx = (\sin t) / (1 - \cos t)$.
- **Differentiate dy/dx using Quotient Rule:**
- • $d/dt [(\sin t) / (1 - \cos t)] = [\cos t(1 - \cos t) - \sin t(\sin t)] / (1 - \cos t)^2$.
- • $= [\cos t - \cos^2 t - \sin^2 t] / (1 - \cos t)^2 = [\cos t - 1] / (1 - \cos t)^2 = -(1 - \cos t) / (1 - \cos t)^2 = -1 / (1 - \cos t)$.
- **Apply Second Derivative Formula:**
- • $d^2y/dx^2 = [-1 / (1 - \cos t)] / (1 - \cos t) = -1 / (1 - \cos t)^2$.
- **Evaluate at $t = \pi / 2$:**
- • $\cos(\pi / 2) = 0$.
- • $d^2y/dx^2 = -1 / (1 - 0)^2 = -1 / 1 = -1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by 2 assuming $\cos(\pi/2) = 1/2$

Question 20 Analysis

★★★ IMPORTANT

Correct Answer: (D) 1 / 3

STEP-BY-STEP SOLUTION

- **Find First Derivatives:**
- • $dx/dt = 6t^2$.
- • $dy/dt = 12t^3$.
- • $dy/dx = (12t^3) / (6t^2) = 2t$.
- **Differentiate dy/dx with respect to t:**
- • $d/dt [2t] = 2$.
- **Compute Second Derivative:**
- • $d^2y/dx^2 = [d/dt (dy/dx)] / (dx/dt) = 2 / (6t^2) = 1 / (3t^2)$.
- **Evaluate at t = 1:**
- • $d^2y/dx^2 = 1 / (3(1)^2) = 1 / 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $1 / (6t^2)$ instead of $2 / (6t^2)$: 1/6.

✗ **(B)** Forgets the 1/2 factor in 2 / 6: 2/3.

✗ **(C)** Neglects to divide by dx/dt = 6: reports 2.

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 9 / 2

STEP-BY-STEP SOLUTION

- **Recall Parametric Arc Length Formula:** $L = \int_a^b \sqrt{([x'(t)]^2 + [y'(t)]^2)}\, dt$.
- **Differentiate x(t) using Product Rule:**
- • $x'(t) = -\sin t + (1 \cdot \sin t + t \cos t) = t \cos t$.
- **Differentiate y(t) using Product Rule:**
- • $y'(t) = \cos t - (1 \cdot \cos t - t \sin t) = t \sin t$.
- **Compute the Integrand:**
- • $[x'(t)]^2 + [y'(t)]^2 = (t \cos t)^2 + (t \sin t)^2 = t^2 \cos^2 t + t^2 \sin^2 t = t^2(\cos^2 t + \sin^2 t) = t^2$.
- • $\sqrt{([x'(t)]^2 + [y'(t)]^2)} = \sqrt{(t^2)} = t$ (since $t \geq 0$).
- **Evaluate the Definite Integral from t = 0 to t = 3:**
- • $L = \int_0^3 t\, dt = [t^2 / 2]_0^3 = 3^2 / 2 - 0 = 9 / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by 2 in the antiderivative of t: $3^2 = 9$.

✗ **(C)** Integrates 1 instead of t: 3.

✗ **(D)** Doubles the correct length: $9/2 \cdot 2 = 9$ or calculation error yielding 6.

Question 22 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) √2 (e^π – 1)

STEP-BY-STEP SOLUTION

- **Differentiate Components:**
- • $x'(t) = e^t \cos t - e^t \sin t = e^t(\cos t - \sin t)$.
- • $y'(t) = e^t \sin t + e^t \cos t = e^t(\sin t + \cos t)$.
- **Compute Sum of Squares:**
- • $[x'(t)]^2 = e^{2t}(\cos^2 t - 2 \sin t \cos t + \sin^2 t) = e^{2t}(1 - 2 \sin t \cos t)$.
- • $[y'(t)]^2 = e^{2t}(\sin^2 t + 2 \sin t \cos t + \cos^2 t) = e^{2t}(1 + 2 \sin t \cos t)$.
- • $[x'(t)]^2 + [y'(t)]^2 = e^{2t}[(1 - 2 \sin t \cos t) + (1 + 2 \sin t \cos t)] = 2e^{2t}$.
- **Take the Square Root:**
- • $\sqrt{(2e^{2t})} = \sqrt{2}\, e^t$.
- **Evaluate Arc Length Integral from t = 0 to t = π:**
- • $L = \int_0^{\pi} \sqrt{2}\, e^t \, dt = \sqrt{2}\, [e^t]_0^{\pi} = \sqrt{2}\,(e^{\pi} - e^0) = \sqrt{2}\,(e^{\pi} - 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the constant factor √2: e^π – 1.

✗ **(C)** Uses factor of 2 instead of √2: 2(e^π – 1).

Question 23 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 3 / 2

STEP-BY-STEP SOLUTION

- **Differentiate Components using Chain Rule:**
- • $x'(t) = 3 \cos^2 t\,(-\sin t) = -3 \cos^2 t \sin t$.
- • $y'(t) = 3 \sin^2 t \cos t$.
- **Compute $[x'(t)]^2 + [y'(t)]^2$:**
- • $[x'(t)]^2 = 9 \cos^4 t \sin^2 t$.
- • $[y'(t)]^2 = 9 \sin^4 t \cos^2 t$.
- • $[x'(t)]^2 + [y'(t)]^2 = 9 \sin^2 t \cos^2 t\,(\cos^2 t + \sin^2 t) = 9 \sin^2 t \cos^2 t$.
- **Take Square Root on [0, π/2]:**
- • $\sqrt{(9 \sin^2 t \cos^2 t)} = 3 \sin t \cos t$ (since $\sin t, \cos t \geq 0$ on $[0, \pi/2]$).
- **Evaluate the Definite Integral:**
- • $L = \int_0^{\pi/2} 3 \sin t \cos t \, dt = 3\,[(1/2) \sin^2 t\,]_0^{(\pi/2)} = (3/2)[\sin^2(\pi/2) - \sin^2(0)\,] = (3/2)(1 - 0) = 3 / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Halves the correct result: 3 / 4.

✗ **(B)** Computes total length of two quadrants: 3.

Question 24 Analysis

★★★★ MANDATORY

Correct Answer: (D) $\int_1^5 \sqrt{((3t^2 - 2)^2 + 64t^2)}\, dt$

STEP-BY-STEP SOLUTION

- **Compute Component Derivatives:**
- • $dx/dt = 3t^2 - 2$.
- • $dy/dt = 8t$.
- **Square the Derivatives:**
- • $[dx/dt]^2 = (3t^2 - 2)^2$.
- • $[dy/dt]^2 = (8t)^2 = 64t^2$.
- **Set Up the Arc Length Integral:**
- • $L = \int_1^5 \sqrt{([dx/dt]^2 + [dy/dt]^2)}\, dt = \int_1^5 \sqrt{((3t^2 - 2)^2 + 64t^2)}\, dt$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Squares original coordinate functions instead of their derivatives.

✗ **(B)** Squares 4t instead of 8t: $16t^2$.

✗ **(C)** Omits square root and squares, simply adding derivatives.

Question 25 Analysis

★★★★ MANDATORY

Correct Answer: (A) 14

STEP-BY-STEP SOLUTION

- **Differentiate Components:**
- • $dx/dt = 6t$.
- • $dy/dt = 6t^2$.
- **Compute the Integrand:**
- • $[dx/dt]^2 + [dy/dt]^2 = (6t)^2 + (6t^2)^2 = 36t^2 + 36t^4 = 36t^2(1 + t^2)$.
- • $\sqrt{(36t^2(1 + t^2))} = 6t\sqrt{(1 + t^2)}$ (for $t \geq 0$).
- **Evaluate the Integral using Substitution:**
- • Let $u = 1 + t^2 \Rightarrow du = 2t\,dt \Rightarrow 6t\,dt = 3\,du$.
- • At $t = 0$: $u = 1 + 0 = 1$.
- • At $t = \sqrt{3}$: $u = 1 + (\sqrt{3})^2 = 1 + 3 = 4$.
- • $L = \int_1^4 3u^{(1/2)}\,du = 3[(2/3)u^{(3/2)}]_1^4 = 2[u^{(3/2)}]_1^4$.
- • At $u = 4$: $4^{(3/2)} = 8$.
- • At $u = 1$: $1^{(3/2)} = 1$.
- • $L = 2(8 - 1) = 2(7) = 14$.

Question 26 Analysis

★★★ IMPORTANT

Correct Answer: (B) 3π

STEP-BY-STEP SOLUTION

- **Compute Derivatives:**
- • $dx/dt = 2\cos t$.
- • $dy/dt = -2\sin t$.
- **Compute Particle Speed:**
- • Speed $= \sqrt{((2\cos t)^2 + (-2\sin t)^2)} = \sqrt{(4\cos^2 t + 4\sin^2 t)} = \sqrt{4} = 2$.
- **Integrate Speed Over Time Interval [0, 3π/2]:**
- • Distance $= \int_0^{3\pi/2} 2\,dt = 2 \cdot (3\pi/2 - 0) = 3\pi$.
- **Geometric Check:** Circle of radius $r = 2$ has circumference $2\pi(2) = 4\pi$. Three-quarters of the circle has length $(3/4)(4\pi) = 3\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses radius 3 or multiplies by 2: 6π.
✗ **(C)** Forgets the radius $r = 2$: $3\pi / 2$.
✗ **(D)** Calculates the full circumference of the circle: 4π.

Question 27 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 12

STEP-BY-STEP SOLUTION

- **Distance Traveled Formula:** Distance $= \int_0^3 \sqrt{([x'(t)]^2 + [y'(t)]^2)}\,dt$.
- **Simplify the Sum of Squares:**
- • $[x'(t)]^2 + [y'(t)]^2 = (t^2 - 1)^2 + (2t)^2 = (t^4 - 2t^2 + 1) + 4t^2 = t^4 + 2t^2 + 1$.
- • Notice this is a perfect square: $t^4 + 2t^2 + 1 = (t^2 + 1)^2$.
- **Take Square Root:**
- • $\sqrt{((t^2 + 1)^2)} = t^2 + 1$.
- **Evaluate the Definite Integral:**
- • Distance $= \int_0^3 (t^2 + 1)\,dt = [(1/3)t^3 + t]_0^3$.
- • At $t = 3$: $(1/3)(27) + 3 = 9 + 3 = 12$.
- • At $t = 0$: 0.
- • Total Distance $= 12$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates net displacement instead of distance or arithmetic mistake.
✗ **(B)** Calculates only the cubic term: $(1/3)(27) = 9$, omitting the linear term.
✗ **(D)** Calculation error: 15.

Question 28 Analysis

★★★★ MANDATORY

Correct Answer: (D) 17.5

STEP-BY-STEP SOLUTION

- **Compute Speed $v(t) = \sqrt{([x'(t)]^2 + [y'(t)]^2)}$ at Each Time:**
- • At $t = 0$: $v(0) = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.
- • At $t = 1$: $v(1) = \sqrt{(4^2 + 3^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$.
- • At $t = 2$: $v(2) = \sqrt{(0^2 + 5^2)} = \sqrt{25} = 5$.
- • At $t = 3$: $v(3) = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.
- **Apply Trapezoidal Rule with $\Delta t = 1$:**
- • Distance $\approx (\Delta t / 2) \cdot [\, v(0) + 2v(1) + 2v(2) + v(3) \,]$.
- • $(1 / 2) \cdot [\, 5 + 2(5) + 2(5) + 10 \,] = (1 / 2) \cdot [\, 5 + 10 + 10 + 10 \,] = (1 / 2) \cdot 35 = 17.5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a Right Riemann sum: $1 \cdot (5 + 5 + 10) = 20.0$.
✗ **(B)** Forgets to divide by 2 in the trapezoidal rule: 35.0.
✗ **(C)** Uses a Left Riemann sum: $1 \cdot (5 + 5 + 5) = 15.0$.

Question 29 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\ln(\sqrt{2} + 1)$

STEP-BY-STEP SOLUTION

- **Differentiate Components with respect to t:**
- • $dx/dt = (1 / \sec t) \cdot (\sec t \tan t) = \tan t$.
- • $dy/dt = 1$.
- **Compute the Integrand:**
- • $[dx/dt]^2 + [dy/dt]^2 = \tan^2 t + 1^2 = \sec^2 t$.
- • $\sqrt{(\sec^2 t)} = \sec t$ (since $\sec t > 0$ on $[0, \pi/4]$).
- **Evaluate the Arc Length Integral:**
- • $L = \int_0^{\pi/4} \sec t \, dt = [\, \ln|\sec t + \tan t| \,]_0^{(\pi/4)}$.
- • At $t = \pi/4$: $\sec(\pi/4) = \sqrt{2}$, $\tan(\pi/4) = 1 \Rightarrow \ln(\sqrt{2} + 1)$.
- • At $t = 0$: $\sec(0) = 1$, $\tan(0) = 0 \Rightarrow \ln(1 + 0) = \ln(1) = 0$.
- • $L = \ln(\sqrt{2} + 1) - 0 = \ln(\sqrt{2} + 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in tangent argument: $\ln(\sqrt{2} - 1)$.
✗ **(C)** Omits the natural logarithm in antiderivative of secant: $\sqrt{2} + 1$

Question 30 Analysis

★★★ IMPORTANT

Correct Answer: (B) 5

STEP-BY-STEP SOLUTION

- **Find Component Derivatives:**
- • $dx/dt = 6$.
- • $dy/dt = 8$.
- **Compute Speed:**
- • Speed $= \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.
- **Set Up Arc Length Equation:**
- • Length $= \int_0^k 10 \, dt = 10k$.
- **Solve for k:**
- • $10k = 50 \Rightarrow k = 5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides 50 by 5 instead of 10: 10.
✗ **(C)** Halves 50: 25.
✗ **(D)** Assumes k equals the length: 50.

Question 31 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) ⟨ 12, 4 ⟩

STEP-BY-STEP SOLUTION

- **Find the Velocity Vector v(t) = r'(t):**
- • $v(t) = \langle d/dt(t^3 - 3t), d/dt(2t^2) \rangle = \langle 3t^2 - 3, 4t \rangle$.
- **Find the Acceleration Vector a(t) = v'(t) = r''(t):**
- • $a(t) = \langle d/dt(3t^2 - 3), d/dt(4t) \rangle = \langle 6t, 4 \rangle$.
- **Evaluate at t = 2:**
- • $a(2) = \langle 6(2), 4 \rangle = \langle 12, 4 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates at t = 1: ⟨ 6, 4 ⟩.
✗ **(B)** Reports the velocity vector $v(2) = \langle 3(4) - 3, 4(2) \rangle = \langle 9, 8 \rangle$ instead of acceleration.
✗ **(D)** Differentiates $2t^2$ as $4t^2$: ⟨ 12, 8 ⟩.

Question 32 Analysis

★★★★ MANDATORY

Correct Answer: (D) 8

STEP-BY-STEP SOLUTION

- **Compute Velocity Vector v(t) using Chain Rule:**
- • $x'(t) = 4(-\sin(2t)) \cdot 2 = -8\sin(2t)$.
- • $y'(t) = 4(\cos(2t)) \cdot 2 = 8\cos(2t)$.
- • $v(t) = \langle -8\sin(2t), 8\cos(2t) \rangle$.
- **Compute Speed ||v(t)||:**
- • $\text{Speed} = \sqrt{((-8\sin(2t))^2 + (8\cos(2t))^2)} = \sqrt{(64\sin^2(2t) + 64\cos^2(2t))}$.
- • $\sqrt{(64(\sin^2(2t) + \cos^2(2t)))} = \sqrt{64} = 8$ (constant speed for uniform circular motion).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports only one component of the velocity vector.
✗ **(B)** Forgets the chain rule factor of 2: $\sqrt{16} = 4$.
✗ **(C)** Squares speed: 64 or 16.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\sqrt{5}$

STEP-BY-STEP SOLUTION

- **Find Velocity Components:**
- • $x'(t) = d/dt[e^{2t}] = 2e^{2t}$.
- • $y'(t) = d/dt[\ln(t + 1)] = 1/(t + 1)$.
- **Evaluate Velocity at t = 0:**
- • $x'(0) = 2e^0 = 2(1) = 2$.
- • $y'(0) = 1/(0 + 1) = 1$.
- • Velocity vector at t = 0 is $v(0) = \langle 2, 1 \rangle$.
- **Compute Speed at t = 0:**
- • $\text{Speed} = ||v(0)|| = \sqrt{((x'(0))^2 + (y'(0))^2)} = \sqrt{(2^2 + 1^2)} = \sqrt{(4 + 1)} = \sqrt{5}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds components directly without squaring: 2 + 1 = 3.
✗ **(C)** Forgets chain rule factor 2 on e^{2t}: $\sqrt{(1 + 1)} = \sqrt{2}$, or calculation error giving $\sqrt{3}$.
✗ **(D)** Squares without taking square root: 4 + 1 = 5.

Question 34 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 0.6

STEP-BY-STEP SOLUTION

- **Condition for Orthogonal Vectors:**
- • Two non-zero vectors v and a are orthogonal if and only if their dot product equals zero: $v(t) \cdot a(t) = 0$.
- **Find Velocity Vector:**
- • $v(t) = \langle d/dt(t^2 - 6t), d/dt(2t^2) \rangle = \langle 2t - 6, 4t \rangle$.
- **Find Acceleration Vector:**
- • $a(t) = \langle d/dt(2t - 6), d/dt(4t) \rangle = \langle 2, 4 \rangle$.
- **Compute Dot Product v(t) · a(t):**
- • $v(t) \cdot a(t) = (2t - 6)(2) + (4t)(4) = 4t - 12 + 16t = 20t - 12$.
- **Solve 20t – 12 = 0:**
- • $20t = 12 \Rightarrow t = 12 / 20 = 3 / 5 = 0.6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Solves $2t - 6 = 0$ forgetting the second component: $2t = 6 \Rightarrow t = 3 / 2 = 1.5$.

✗ **(C)** Solves where x-velocity is zero: $2t - 6 = 0 \Rightarrow t = 3.0$.

✗ **(D)** Arithmetic error: $12 / 10 = 1.2$.

Question 35 Analysis

★★★★ MANDATORY

Correct Answer: (C) π / 3

STEP-BY-STEP SOLUTION

- **Recall Direction Angle Formula:**
- • The direction angle θ of velocity vector $\langle v_x, v_y \rangle$ satisfies $\tan \theta = v_y / v_x$.
- **Given Velocity Components:**
- • $v_x = 2 > 0$ and $v_y = 2\sqrt{3} > 0$ (First Quadrant).
- • $\tan \theta = (2\sqrt{3}) / 2 = \sqrt{3}$.
- **Determine Angle θ in [0, 2π):**
- • $\tan \theta = \sqrt{3}$ in the first quadrant gives $\theta = \pi / 3$ (or 60°).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Places vector in the second quadrant: $2\pi / 3$.

✗ **(B)** Inverts the quotient: $\tan \theta = 2 / (2\sqrt{3}) = 1/\sqrt{3} \Rightarrow \theta = \pi / 6$.

✗ **(D)** Assumes components are equal: $\pi / 4$.

Question 36 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) t > 1

STEP-BY-STEP SOLUTION

- **Condition for Increasing Speed:**
- • Speed $||v(t)||$ is increasing when $d/dt\, ||v(t)|| > 0$, which is equivalent to $v(t) \cdot a(t) > 0$.
- **Compute Velocity and Acceleration:**
- • $v(t) = \langle 2t - 2, 3 \rangle$.
- • $a(t) = \langle 2, 0 \rangle$.
- **Compute Dot Product v(t) · a(t):**
- • $v(t) \cdot a(t) = (2t - 2)(2) + (3)(0) = 4t - 4 = 4(t - 1)$.
- **Determine When v(t) · a(t) > 0:**
- • $4(t - 1) > 0 \Rightarrow t - 1 > 0 \Rightarrow t > 1$.
- • Thus, the particle's speed is increasing for $t > 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes speed is always increasing for $t > 0$ because $y'(t) = 3 > 0$.

✗ **(B)** Identifies where speed is decreasing ($t < 1$).

✗ **(C)** Solves where x-component of position is zero: $t(t - 2) = 0 \Rightarrow t = 2$.

Question 37 Analysis

★★★★ MANDATORY

Correct Answer: (A) Decreasing, because v(4) · a(4) < 0

STEP-BY-STEP SOLUTION

- **Rate of Change of Speed:**
- • $d/dt\,||v(t)|| = (v(t) \cdot a(t)) / ||v(t)||$.
- • Since $||v(t)|| > 0$, the sign of the derivative of speed is determined by the dot product $v(t) \cdot a(t)$.
- **Compute the Dot Product at t = 4:**
- • $v(4) \cdot a(4) = (-3)(2) + (4)(-1) = -6 - 4 = -10$.
- **Conclusion:**
- • Since $v(4) \cdot a(4) = -10 < 0$, the particle's speed is decreasing at time $t = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Incorrectly computes dot product with sign error: $-6 + 4 = -2$ or $6 - 4 = 2 > 0$.
✗ **(C)** Claims vectors are orthogonal without calculating dot product.
✗ **(D)** Confuses positive speed with increasing speed.

Question 38 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) t = 2 only

STEP-BY-STEP SOLUTION

- **Condition for Particle at Rest:**
- • A particle moving in two dimensions is at rest if and only if its velocity vector is the zero vector: $v(t) = \langle 0, 0 \rangle$.
- • This requires BOTH $x'(t) = 0$ and $y'(t) = 0$ simultaneously.
- **Solve x'(t) = 0:**
- • $3t^2 - 12 = 0 \Rightarrow 3t^2 = 12 \Rightarrow t^2 = 4 \Rightarrow t = 2$ or $t = -2$.
- **Solve y'(t) = 0:**
- • $2t - 4 = 0 \Rightarrow 2t = 4 \Rightarrow t = 2$.
- **Find Common Solutions:**
- • The only value of t satisfying both equations simultaneously is $t = 2$.
- • (At $t = -2$, $y'(-2) = -8 \neq 0$, so the particle is moving vertically downward, not at rest.)

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes $t = -2$ where only the horizontal velocity is zero.
✗ **(C)** Chooses the negative root.
✗ **(D)** Claims particle is never at rest by confusing vector rest with single-axis motion.

Question 39 Analysis

★★★ IMPORTANT

Correct Answer: (C) 9

STEP-BY-STEP SOLUTION

- **Compute Velocity Vector v(t):**
- • $v(t) = \langle 3\cos(3t), -3\sin(3t) \rangle$.
- **Compute Acceleration Vector a(t):**
- • $a(t) = \langle -9\sin(3t), -9\cos(3t) \rangle$.
- **Compute Magnitude of Acceleration ||a(t)||:**
- • $||a(t)|| = \sqrt{((-9\sin(3t))^2 + (-9\cos(3t))^2)} = \sqrt{(81\sin^2(3t) + 81\cos^2(3t))}$.
- • $\sqrt{(81(\sin^2(3t) + \cos^2(3t)))} = \sqrt{81} = 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits chain rule factors: $||a(t)|| = 1$.
✗ **(B)** Reports speed $||v(t)|| = 3$ instead of acceleration magnitude.
✗ **(D)** Multiplies by another factor of 3: 27.

Question 40 Analysis

★★★★ MANDATORY

Correct Answer: (D) ⟨ √2 / 2, √2 / 2 ⟩

STEP-BY-STEP SOLUTION

- **Recall Unit Tangent Vector Definition:**
- • $T(t) = v(t) / ||v(t)||$.
- **Find Velocity Vector at t = 1:**
- • $v(t) = \langle 2t, 2 \rangle$.
- • At $t = 1$: $v(1) = \langle 2(1), 2 \rangle = \langle 2, 2 \rangle$.
- **Compute Magnitude $||v(1)||$:**
- • $||v(1)|| = \sqrt{(2^2 + 2^2)} = \sqrt{(4 + 4)} = \sqrt{8} = 2\sqrt{2}$.
- **Divide Vector by Its Magnitude:**
- • $T(1) = \langle 2 / (2\sqrt{2}), 2 / (2\sqrt{2}) \rangle = \langle 1 / \sqrt{2}, 1 / \sqrt{2} \rangle = \langle \sqrt{2} / 2, \sqrt{2} / 2 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by 2: $\langle 1, 1 \rangle$, which has magnitude $\sqrt{2} \neq 1$.

✗ **(B)** Reports the non-normalized velocity vector $v(1) = \langle 2, 2 \rangle$.

✗ **(C)** Divides by 4 instead of $2\sqrt{2}$: $\langle 1/2, 1/2 \rangle$.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) ⟨ 28, 21 ⟩

STEP-BY-STEP SOLUTION

- **Fundamental Theorem of Calculus for Vectors:**
- • $r(3) = r(1) + \int_1^3 v(t)\,dt = \langle x(1) + \int_1^3 3t^2\,dt, y(1) + \int_1^3 4t\,dt \rangle$.
- **Compute the x-component:**
- • $\int_1^3 3t^2\,dt = [t^3]_1^3 = 3^3 - 1^3 = 27 - 1 = 26$.
- • $x(3) = 2 + 26 = 28$.
- **Compute the y-component:**
- • $\int_1^3 4t\,dt = [2t^2]_1^3 = 2(3)^2 - 2(1)^2 = 18 - 2 = 16$.
- • $y(3) = 5 + 16 = 21$.
- **Combine Components:**
- • $r(3) = \langle 28, 21 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Reports only the displacement vector $\langle 26, 16 \rangle$ without adding the initial position $r(1) = \langle 2, 5 \rangle$.

✗ **(C)** Integrates from $t = 0$ instead of $t = 1$: $[t^3]_0^3 = 27$, $[2t^2]_0^3 = 18$.

✗ **(D)** Adds r(1) twice or calculation mistake: $\langle 30, 26 \rangle$.

Question 42 Analysis

★★★★ MANDATORY

Correct Answer: (B) ⟨ 7, 13 ⟩

STEP-BY-STEP SOLUTION

- **Velocity from Acceleration:**
- • $v(2) = v(0) + \int_0^2 a(t)\,dt$.
- **Compute Component Integrals:**
- • $v_x(2) = 3 + \int_0^2 2\,dt = 3 + [2t]_0^2 = 3 + 4 = 7$.
- • $v_y(2) = 1 + \int_0^2 6t\,dt = 1 + [3t^2]_0^2 = 1 + 3(4) = 1 + 12 = 13$.
- **Combine Components:**
- • $v(2) = \langle 7, 13 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates only the change in velocity $\langle 4, 12 \rangle$ without adding $v(0) = \langle 3, 1 \rangle$.

✗ **(C)** Integrates $6t$ to $6t^2$ instead of $3t^2$: $1 + 24 = 25$.

✗ **(D)** Adds initial velocity incorrectly.

Question 43 Analysis

★★★ IMPORTANT

Correct Answer: (C) ⟨ 0, 2 ⟩

STEP-BY-STEP SOLUTION

- **Displacement Vector Definition:**
- • Displacement = $\int_0^{\pi} v(t)\,dt = \langle \int_0^{\pi} \cos t\,dt,\ \int_0^{\pi} \sin t\,dt \rangle$.
- **Integrate Each Component:**
- • x-component: $[\sin t]_0^{\pi} = \sin(\pi) - \sin(0) = 0 - 0 = 0$.
- • y-component: $[-\cos t]_0^{\pi} = -\cos(\pi) - (-\cos(0)) = -(-1) - (-1) = 1 + 1 = 2$.
- **Combine Components:**
- • Displacement = ⟨ 0, 2 ⟩.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses components or sign in evaluating cosine: ⟨ −2, 0 ⟩.
✗ **(B)** Believes both components cancel to 0 over the interval $[0, \pi]$.
✗ **(D)** Sign error in evaluating $-\cos(\pi) - (-\cos(0))$, obtaining 1.

Question 44 Analysis

★★★★ MANDATORY

Correct Answer: (D) ⟨ 3, −3 + (ln 3)²⟩

STEP-BY-STEP SOLUTION

- **Compute Position Components via FTC:**
- • $x(\ln 3) = x(0) + \int_0^{\ln 3} e^t\,dt = 1 + [e^t]_0^{\ln 3} = 1 + (e^{\ln 3} - e^0) = 1 + (3 - 1) = 3$.
- • $y(\ln 3) = y(0) + \int_0^{\ln 3} 2t\,dt = -3 + [t^2]_0^{\ln 3} = -3 + (\ln 3)^2$.
- **Combine Coordinates:**
- • $r(\ln 3) = \langle 3, -3 + (\ln 3)^2 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates 2t as 2 ln 3 instead of $(\ln 3)^2$.
✗ **(B)** Forgets to add initial position x(0) = 1: 3 − 1 = 2.
✗ **(C)** Forgets initial position y(0) = −3 in y-component.

Question 45 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 4

STEP-BY-STEP SOLUTION

- **Integrate Horizontal Velocity to find x(t):**
- • $x(t) = x(0) + \int_0^t (1/(u+1))\,du = 0 + [\ln(u+1)]_0^t = \ln(t+1) - \ln(1) = \ln(t+1)$.
- **Set x(t) = ln 5:**
- • $\ln(t+1) = \ln 5$.
- • $t + 1 = 5 \Rightarrow t = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sets t = 5 forgetting the +1 term inside the logarithm.
✗ **(C)** Exponentiates with base e incorrectly: $e^5 - 1$.
✗ **(D)** Takes natural logarithm of the root: ln 4.

Question 46 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 144 ft

STEP-BY-STEP SOLUTION

- **Find Vertical Velocity Component v_y(t):**
- • $v_y(t) = v_y(0) + \int_0^t (-32)\,du = 64 - 32t$.
- **Condition for Maximum Height:**
- • Maximum height occurs when vertical velocity equals zero: $v_y(t) = 0$.
- • $64 - 32t = 0 \Rightarrow 32t = 64 \Rightarrow t = 2$ seconds.
- **Compute Vertical Position y(2):**
- • $y(2) = y(0) + \int_0^2 v_y(t)\,dt = 80 + \int_0^2 (64 - 32t)\,dt$.
- • $[64t - 16t^2]_0^2 = 64(2) - 16(4) = 128 - 64 = 64$ ft (change in height).
- • $y(2) = 80 + 64 = 144$ ft.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports only the change in height (64 ft) without adding the initial launch height $y(0) = 80$ ft.
✗ **(C)** Neglects the $-16t^2$ term: $80 + 48 = 128$ ft.
✗ **(D)** Calculation error: 160 ft.

Question 47 Analysis

★★★★ MANDATORY

Correct Answer: (C) ⟨ 19, 7.5 ⟩

STEP-BY-STEP SOLUTION

- **Trapezoidal Sum for x-displacement (Δt = 1):**
- • $\Delta x \approx (1/2) \cdot [x'(0) + 2x'(1) + 2x'(2) + x'(3)]$.
- • $(1/2) \cdot [2 + 2(5) + 2(7) + 4] = (1/2) \cdot [2 + 10 + 14 + 4] = (1/2) \cdot 30 = 15$.
- • Position $x(3) \approx x(0) + \Delta x = 4 + 15 = 19$.
- **Trapezoidal Sum for y-displacement (Δt = 1):**
- • $\Delta y \approx (1/2) \cdot [y'(0) + 2y'(1) + 2y'(2) + y'(3)]$.
- • $(1/2) \cdot [1 + 2(3) + 2(4) + 2] = (1/2) \cdot [1 + 6 + 8 + 2] = (1/2) \cdot 17 = 8.5$.
- • Position $y(3) \approx y(0) + \Delta y = -1 + 8.5 = 7.5$.
- **Combine Coordinates:**
- • $r(3) \approx \langle 19, 7.5 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculation error in trapezoidal sums.
✗ **(B)** Reports displacement vector ⟨ 15, 8.5 ⟩ directly without adding initial position $r(0) = \langle 4, -1 \rangle$.
✗ **(D)** Adds Δy to 0 instead of −1: ⟨ 19, 8.5 ⟩.

Question 48 Analysis

★★★★ MANDATORY

Correct Answer: (D) ⟨ 18, 6 ⟩

STEP-BY-STEP SOLUTION

- **Compute x-position via FTC:**
- • $x(4) = x(1) + \int_1^4 2t\,dt = 3 + [t^2]_1^4 = 3 + (16 - 1) = 3 + 15 = 18$.
- **Compute y-position via FTC:**
- • $y(4) = y(1) + \int_1^4 t^{-1/2}\,dt = 4 + [2\sqrt{t}]_1^4$.
- • $[2\sqrt{t}]_1^4 = 2\sqrt{4} - 2\sqrt{1} = 2(2) - 2(1) = 4 - 2 = 2$.
- • $y(4) = 4 + 2 = 6$.
- **Combine Coordinates:**
- • $r(4) = \langle 18, 6 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects subtracting 1 in $[t^2]_1^4$: $3 + 13 = 16$.
✗ **(B)** Reports displacement vector ⟨ 15, 2 ⟩ without adding initial position ⟨ 3, 4 ⟩.
✗ **(C)** Evaluates upper limit $2\sqrt{4} = 4$ without subtracting lower limit $2\sqrt{1} = 2$: $4 + 4 = 8$.

Question 49 Analysis

★★★ IMPORTANT

Correct Answer: (A) 20

STEP-BY-STEP SOLUTION

- **Find Position Vector r(t) by Integration:**
- • $x(2) = \int_0^2 6t\,dt = [\,3t^2\,]_0^2 = 3(4) = 12$.
- • $y(2) = \int_0^2 8t\,dt = [\,4t^2\,]_0^2 = 4(4) = 16$.
- • Position at t = 2 is $r(2) = \langle 12, 16 \rangle$.
- **Compute Distance from Origin:**
- • Distance $= \sqrt{(12^2 + 16^2)} = \sqrt{(144 + 256)} = \sqrt{400} = 20$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates speed at t = 1: $\sqrt{(6^2 + 8^2)} = 10$.

✗ **(C)** Evaluates speed at t = 2: $||v(2)|| = \sqrt{(12^2 + 16^2)}$? No, $v(2) = \langle 12, 16 \rangle \Rightarrow$ speed is 20, but calculation error gives 40.

✗ **(D)** Adds coordinates directly: 12 + 16 = 28.

Question 50 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) ⟨ 0, 0 ⟩

STEP-BY-STEP SOLUTION

- **Compute Horizontal Position x(2):**
- • $x(2) = 0 + \int_0^2 \sin(\pi t)\,dt = [\,-(1/\pi)\cos(\pi t)\,]_0^2$.
- • At t = 2: $-(1/\pi)\cos(2\pi) = -1/\pi$.
- • At t = 0: $-(1/\pi)\cos(0) = -1/\pi$.
- • $x(2) = -1/\pi - (-1/\pi) = 0$.
- **Compute Vertical Position y(2):**
- • $y(2) = 0 + \int_0^2 \cos(\pi t)\,dt = [\,(1/\pi)\sin(\pi t)\,]_0^2$.
- • At t = 2: $(1/\pi)\sin(2\pi) = 0$.
- • At t = 0: $(1/\pi)\sin(0) = 0$.
- • $y(2) = 0 - 0 = 0$.
- **Combine Coordinates:**
- • $r(2) = \langle 0, 0 \rangle$ (the particle completes one full period and returns to the origin).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates over half a period [0, 1] for x: 2/π

Question 51 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) Their paths intersect, but the particles do not collide

STEP-BY-STEP SOLUTION

- **Collision Requirement:**
- • Particles collide if and only if they occupy the same position at the SAME time t: $r_A(t) = r_B(t)$.
- • Equate y-coordinates: $3t - 2 = 2t + 2 \Rightarrow t = 4$.
- • Check x-coordinates at t = 4: $x_A(4) = 2(4) + 1 = 9$, while $x_B(4) = 4^2 - 2 = 16 - 2 = 14$.
- • Since $x_A(4) \neq x_B(4)$, the particles do NOT collide.
- **Path Intersection Requirement:**
- • Paths intersect if the curves cross at some point (x, y), possibly at DIFFERENT times t_1 and t_2.
- • From Particle A: $x = 2t_1 + 1 \Rightarrow t_1 = (x - 1)/2$. Then $y = 3((x - 1)/2) - 2 = (3x - 7)/2$.
- • Test on Particle B: At $t_2 = 3$, $x_B(3) = 3^2 - 2 = 7$ and $y_B(3) = 2(3) + 2 = 8$.
- • Plug x = 7 into Path A: $y = (3(7) - 7)/2 = 14/2 = 7 \neq 8$.
- • Let's find common point: $x_B(t_2) = t_2^2 - 2$, $y_B(t_2) = 2t_2 + 2$. On Path A: $2t_2 + 2 = (3(t_2^2 - 2) - 7)/2 \Rightarrow 4t_2 + 4 = 3t_2^2 - 13 \Rightarrow 3t_2^2 - 4t_2 - 17 = 0$.
- • Discriminant $\Delta = (-4)^2 - 4(3)(-17) = 16 + 204 = 220 > 0$. Positive roots exist!
- • Therefore, the paths cross, but the particles never occupy the intersection at the same time.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculation mistake assuming collision at t = 3.

✗ **(B)** Fails to check x-coordinates at t = 4 where y-coordinates match.

✗ **(D)** Assumes that because particles don't collide, their trajectories can never cross.

Question 52 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\sqrt{(\pi / 2)}$

STEP-BY-STEP SOLUTION

- **Condition for Moving Purely Vertically:**
- • The particle moves purely vertically when its horizontal velocity is zero ($x'(t) = 0$) and vertical velocity is non-zero ($y'(t) \neq 0$).
- **Solve $x'(t) = 0$:**
- • $\cos(t^2) = 0$.
- • For $t > 0$, the first positive root of the cosine function is $\pi / 2$.
- • $t^2 = \pi / 2 \Rightarrow t = \sqrt{(\pi / 2)}$.
- **Verify $y'(t) \neq 0$:**
- • $y'(\sqrt{(\pi / 2)}) = 3\sqrt{(\pi / 2)} \neq 0$.
- • Thus, the particle first moves purely vertically at $t = \sqrt{(\pi / 2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Chooses the second positive root: $t = \sqrt{(3\pi / 2)}$.
✗ **(B)** Solves where $\sin(t^2) = 0$: $t = \sqrt{\pi}$.
✗ **(C)** Forgets the square root: $t = \pi / 2$.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $3\sqrt{5} / 5$

STEP-BY-STEP SOLUTION

- **Distance Squared Function from Origin:**
- • Let $D(t) = [x(t)]^2 + [y(t)]^2 = (t - 2)^2 + (2t - 1)^2$.
- • $D(t) = (t^2 - 4t + 4) + (4t^2 - 4t + 1) = 5t^2 - 8t + 5$.
- **Find Critical Point of $D(t)$:**
- • $D'(t) = 10t - 8$.
- • Set $D'(t) = 0 \Rightarrow 10t = 8 \Rightarrow t = 8 / 10 = 4 / 5 = 0.8$.
- • Since $D''(t) = 10 > 0$, this yields a global minimum.
- **Compute Minimum Distance:**
- • $D(4/5) = 5(4/5)^2 - 8(4/5) + 5 = 5(16/25) - 32/5 + 5 = 16/5 - 32/5 + 25/5 = 9 / 5$.
- • Minimum Distance $= \sqrt{(9 / 5)} = 3 / \sqrt{5} = 3\sqrt{5} / 5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates distance at $t = 1$: $\sqrt{((-1)^2 + 1^2)} = \sqrt{2}$ or at $t = 2$: $\sqrt{(0 + 9)} = 3$.
✗ **(C)** Reports the distance squared or omits the square root: 9/5 or 3/5.
✗ **(D)** Calculation error in minimizing quadratic.

Question 54 Analysis

★★★★ MANDATORY

Correct Answer: (B) $3 < t < 6$

STEP-BY-STEP SOLUTION

- **Direction Conditions:**
- • Moving to the left means horizontal velocity is negative: $x'(t) < 0$.
- • Moving upward means vertical velocity is positive: $y'(t) > 0$.
- **Read Graphs from Figure:**
- • $x'(t)$ crosses the t-axis at $t = 3$ and remains below the axis ($x'(t) < 0$) for $3 < t \leq 6$.
- • $y'(t)$ crosses the t-axis at $t = 2$ and remains above the axis ($y'(t) > 0$) for $2 < t \leq 6$.
- **Find the Intersection of Conditions:**
- • Both $x'(t) < 0$ and $y'(t) > 0$ on the interval $3 < t < 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Identifies where $x'(t) > 0$ and $y'(t) < 0$ (moving right and downward).
✗ **(C)** Overlaps where $x'(t) > 0$ and $y'(t) > 0$.
✗ **(D)** Identifies where $x'(t) > 0$.

Question 55 Analysis

★★★★ MANDATORY

Correct Answer: (C) (4, −16)

STEP-BY-STEP SOLUTION

- **Condition for Lowest Point:**
- • The lowest point occurs where y(t) has a minimum.
- • Find critical points of y(t): $y'(t) = 3t^2 - 12$.
- • Set $y'(t) = 0 \Rightarrow 3t^2 = 12 \Rightarrow t^2 = 4 \Rightarrow t = 2$ (for $t \geq 0$).
- • Verify minimum: $y''(t) = 6t \Rightarrow y''(2) = 12 > 0$ (local and absolute minimum).
- **Compute Coordinates at t = 2:**
- • $x(2) = (2)^2 = 4$.
- • $y(2) = (2)^3 - 12(2) = 8 - 24 = -16$.
- • The lowest point is (4, −16).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the origin is the minimum.
✗ **(B)** Uses t = 2 directly as the x-coordinate: (2, −16).
✗ **(D)** Multiplies t by 6: (12, −16).

Question 56 Analysis

★★★ IMPORTANT

Correct Answer: (D) The acceleration vector is always orthogonal to the velocity vector (v(t) · a(t) = 0)

STEP-BY-STEP SOLUTION

- **Analyze Constant Speed Property:**
- • Speed squared is constant: $||v(t)||^2 = v(t) \cdot v(t) = c^2$.
- **Differentiate Both Sides with respect to t:**
- • $d/dt\,[\,v(t) \cdot v(t)\,] = d/dt\,[\,c^2\,]$.
- • Using the product rule for dot products: $v'(t) \cdot v(t) + v(t) \cdot v'(t) = 0$.
- • $2\,(v(t) \cdot a(t)) = 0 \Rightarrow v(t) \cdot a(t) = 0$.
- **Conclusion:**
- • The acceleration vector is always orthogonal to the velocity vector (e.g. uniform circular motion).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Position and velocity are only orthogonal for circles centered at the origin, not in general.
✗ **(B)** Acceleration is only zero if both speed and direction are constant (straight line).
✗ **(C)** A particle moving in a circle has constant speed but turns continuously.

Question 57 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) t = 0

STEP-BY-STEP SOLUTION

- **Find the Direction Angle θ(t):**
- • $x'(t) = 2$, $y'(t) = 2t$.
- • $\tan\theta = y'(t) / x'(t) = (2t) / 2 = t$.
- • Thus, $\theta(t) = \arctan(t)$.
- **Compute dθ/dt:**
- • $d\theta/dt = d/dt\,[\,\arctan(t)\,] = 1 / (1 + t^2)$.
- **Maximize dθ/dt on t ≥ 0:**
- • Since $1 + t^2$ is minimized at $t = 0$ (where $1 + t^2 = 1$), the fraction $1 / (1 + t^2)$ achieves its absolute maximum of 1 at $t = 0$.
- • For any $t > 0$, $1 / (1 + t^2) < 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates at t = 1 where $d\theta/dt = 1/2$.
✗ **(C)** Evaluates at t = 2 where $d\theta/dt = 1/5$.
✗ **(D)** Confuses with derivative of $1/(1+t^2)$.

Question 58 Analysis

★★★★ MANDATORY

Correct Answer: (B) t = 2

STEP-BY-STEP SOLUTION

- **Formulate Speed Function:**
- • Speed $S(t) = \sqrt{((4t-8)^2 + 3^2)} = \sqrt{((4t-8)^2 + 9)}$.
- **Minimize Speed:**
- • $S(t)$ is minimized when the expression inside the square root, $(4t-8)^2 + 9$, is minimized.
- • Since $(4t-8)^2 \geq 0$ for all t, its absolute minimum occurs when $4t - 8 = 0$.
- • $4t = 8 \Longrightarrow t = 2$.
- • At $t = 2$, the minimum speed is $\sqrt{(0 + 9)} = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates initial speed $\sqrt{(64 + 9)} = \sqrt{73}$.
✗ **(C)** Solves $4t = 16 \Longrightarrow t = 4$.
✗ **(D)** Solves $3t = 8$.

Question 59 Analysis

★★★★ MANDATORY

Correct Answer: (C) 4π

STEP-BY-STEP SOLUTION

- **Compute Velocity and Speed:**
- • $v(t) = \langle -\pi \sin(\pi t), \pi \cos(\pi t) \rangle$.
- • Speed $= \sqrt{((-\pi \sin(\pi t))^2 + (\pi \cos(\pi t))^2)} = \sqrt{(\pi^2(\sin^2(\pi t) + \cos^2(\pi t)))} = \pi$.
- **Integrate Speed Over [0, 4]:**
- • Distance $= \int_0^4 \pi \, dt = \pi \cdot (4 - 0) = 4\pi$.
- **Geometric Check:**
- • The particle traverses a circle of radius 1 (circumference 2π) with period $T = 2\pi / \pi = 2$ seconds.
- • In 4 seconds, it completes 2 full revolutions: $2 \cdot (2\pi) = 4\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct distance: 8π.
✗ **(B)** Computes length of one revolution: 2π.
✗ **(D)** Reports net displacement magnitude: $||r(4) - r(0)|| = ||\langle 1, 0 \rangle - \langle 1, 0 \rangle|| = 0$.

Question 60 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) ⟨ 3, 0 ⟩

STEP-BY-STEP SOLUTION

- **Find Velocity Vector v(t):**
- • The particle is at rest at $t = 0 \Longrightarrow v(0) = \langle 0, 0 \rangle$.
- • $v(t) = \int a(t) \, dt = \langle \int 4 \, dt, \int 12t \, dt \rangle = \langle 4t, 6t^2 \rangle$.
- **Find Position Vector r(t):**
- • $r(0) = \langle 1, -2 \rangle$.
- • $x(1) = 1 + \int_0^1 4t \, dt = 1 + [2t^2]_0^1 = 1 + 2 = 3$.
- • $y(1) = -2 + \int_0^1 6t^2 \, dt = -2 + [2t^3]_0^1 = -2 + 2 = 0$.
- **Combine Coordinates:**
- • $r(1) = \langle 3, 0 \rangle$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to integrate vertical velocity: $\langle 3, -2 \rangle$.
✗ **(B)** Reports displacement vector $\langle 2, 2 \rangle$ without adding initial position $\langle 1, -2 \rangle$.
✗ **(C)** Adds acceleration directly: $\langle 1 + 4, -2 + 4 \rangle = \langle 5, 2 \rangle$.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

- **Recall Polar Derivative Formula:**
- • $dy/dx = (dy/d\theta) / (dx/d\theta)$.
- • Since $x = r\cos\theta$ and $y = r\sin\theta$:
- • $dy/d\theta = (dr/d\theta)\sin\theta + r\cos\theta$.
- • $dx/d\theta = (dr/d\theta)\cos\theta - r\sin\theta$.
- **Evaluate r and $dr/d\theta$ at $\theta = \pi / 2$:**
- • $r = 2 + 2\cos(\pi/2) = 2 + 2(0) = 2$.
- • $dr/d\theta = -2\sin\theta \Longrightarrow$ at $\theta = \pi/2$: $dr/d\theta = -2(1) = -2$.
- **Compute $dy/d\theta$ and $dx/d\theta$ at $\theta = \pi / 2$:**
- • $dy/d\theta = (-2)\sin(\pi/2) + 2\cos(\pi/2) = (-2)(1) + 2(0) = -2$.
- • $dx/d\theta = (-2)\cos(\pi/2) - 2\sin(\pi/2) = (-2)(0) - 2(1) = -2$.
- **Compute Tangent Slope:**
- • $dy/dx = (dy/d\theta) / (dx/d\theta) = -2 / -2 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $-\sqrt{3}$

STEP-BY-STEP SOLUTION

- **Evaluate r and $dr/d\theta$ at $\theta = \pi / 3$:**
- • $\sin(\pi/3) = \sqrt{3} / 2 \Longrightarrow r = 4(\sqrt{3} / 2) = 2\sqrt{3}$.
- • $dr/d\theta = 4\cos\theta \Longrightarrow$ at $\theta = \pi/3$: $dr/d\theta = 4(1/2) = 2$.
- **Compute $dy/d\theta$:**
- • $dy/d\theta = (dr/d\theta)\sin\theta + r\cos\theta = (2)(\sqrt{3} / 2) + (2\sqrt{3})(1/2) = \sqrt{3} + \sqrt{3} = 2\sqrt{3}$.
- **Compute $dx/d\theta$:**
- • $dx/d\theta = (dr/d\theta)\cos\theta - r\sin\theta = (2)(1/2) - (2\sqrt{3})(\sqrt{3} / 2) = 1 - 3 = -2$.
- **Compute Tangent Slope dy/dx:**
- • $dy/dx = (dy/d\theta) / (dx/d\theta) = (2\sqrt{3}) / (-2) = -\sqrt{3}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error dropping the negative sign in $dx/d\theta$: $+\sqrt{3}$.

✗ **(C)** Inverts the fraction: $-1 / \sqrt{3}$.

✗ **(D)** Inverts the fraction and drops negative sign: $1 / \sqrt{3}$.

Question 63 Analysis

★★★★ MANDATORY

Correct Answer: (C) $dr/d\theta = -\sqrt{3}$, meaning the distance between the curve and the origin is decreasing

STEP-BY-STEP SOLUTION

- **Compute $dr/d\theta$:**
- • $dr/d\theta = d/d\theta\,[\,3 - 2\sin\theta\,] = -2\cos\theta$.
- • At $\theta = \pi/6$: $\cos(\pi/6) = \sqrt{3} / 2 \Longrightarrow dr/d\theta = -2(\sqrt{3} / 2) = -\sqrt{3}$.
- **Interpret the Geometric Meaning:**
- • In polar coordinates, r represents the distance from the origin to a point on the curve.
- • Since $dr/d\theta < 0$, as θ increases through $\pi/6$, r is decreasing.
- • Therefore, the distance between the curve and the origin is decreasing at $\theta = \pi/6$.
- • (Note: $dr/d\theta$ is NOT the slope of the tangent line dy/dx!)

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $\sin(\pi/6) = 1/2$ instead of derivative of sine, getting -1.

✗ **(B)** Confuses the radial rate of change $dr/d\theta$ with the Cartesian slope dy/dx.

✗ **(D)** Sign error in differentiating sine.

Question 64 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\theta = \pi / 2$

STEP-BY-STEP SOLUTION

- **Horizontal Tangent Condition:** $dy/d\theta = 0$ with $dx/d\theta \neq 0$.
- **Formulate y in terms of θ:**
- • $y = r \sin\theta = (1 + \sin\theta)\sin\theta = \sin\theta + \sin^2\theta$.
- **Differentiate y with respect to θ:**
- • $dy/d\theta = \cos\theta + 2\sin\theta\cos\theta = \cos\theta(1 + 2\sin\theta)$.
- **Set $dy/d\theta = 0$:**
- • 1. $\cos\theta = 0 \Rightarrow \theta = \pi/2$ or $\theta = 3\pi/2$.
- • 2. $1 + 2\sin\theta = 0 \Rightarrow \sin\theta = -1/2 \Rightarrow \theta = 7\pi/6$ or $\theta = 11\pi/6$.
- **Test $\theta = 3\pi/2$:**
- • At $\theta = 3\pi/2$: $r = 1 + \sin(3\pi/2) = 1 - 1 = 0$ (the pole / cusp). Both $dy/d\theta = 0$ and $dx/d\theta = 0$.
- **Test $\theta = \pi/2$:**
- • $x = (1 + \sin\theta)\cos\theta \Rightarrow dx/d\theta = \cos^2\theta - \sin\theta(1 + \sin\theta) = \cos^2\theta - \sin\theta - \sin^2\theta = \cos(2\theta) - \sin\theta$.
- • At $\theta = \pi/2$: $dx/d\theta = \cos(\pi) - \sin(\pi/2) = -1 - 1 = -2 \neq 0$.
- • Since $dy/d\theta = 0$ and $dx/d\theta = -2 \neq 0$, there is a horizontal tangent at $\theta = \pi/2$.

Question 65 Analysis

★★★★ MANDATORY

Correct Answer: (A) $\theta = 0$

STEP-BY-STEP SOLUTION

- **Convert to Rectangular Coordinates:**
- • $r = 2\cos\theta \Rightarrow r^2 = 2r\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$.
- • This is a circle of radius 1 centered at (1, 0).
- • Vertical tangents occur at the extreme left ($x = 0$) and extreme right ($x = 2$).
- **Check $\theta = 0$:**
- • At $\theta = 0$: $r = 2\cos(0) = 2$. $x = r\cos(0) = 2(1) = 2$, $y = r\sin(0) = 0$.
- • This is the rightmost point (2, 0), where the circle has a vertical tangent!
- **Verify via $dx/d\theta$:**
- • $x = r\cos\theta = 2\cos^2\theta \Rightarrow dx/d\theta = -4\cos\theta\sin\theta = -2\sin(2\theta)$.
- • At $\theta = 0$: $dx/d\theta = -2\sin(0) = 0$.
- • $dy/d\theta$: $y = 2\cos\theta\sin\theta = \sin(2\theta) \Rightarrow dy/d\theta = 2\cos(2\theta)$.
- • At $\theta = 0$: $dy/d\theta = 2\cos(0) = 2 \neq 0$.
- • Since $dx/d\theta = 0$ and $dy/d\theta \neq 0$ at $\theta = 0$, the curve has a vertical tangent at $\theta = 0$.

WHY OTHER OPTIONS ARE INCORRECT

Question 66 Analysis

★★★★ MANDATORY

Correct Answer: (B) The line $\theta = \alpha$

STEP-BY-STEP SOLUTION

- **Theorem for Tangents at the Pole:**
- • If $r = f(\theta)$ equals zero at $\theta = \alpha$ (meaning the curve passes through the pole) and $f'(\alpha) \neq 0$, then the slope of the tangent line is:
- • $dy/dx = (f'(\alpha)\sin\alpha + f(\alpha)\cos\alpha) / (f'(\alpha)\cos\alpha - f(\alpha)\sin\alpha) = (f'(\alpha)\sin\alpha) / (f'(\alpha)\cos\alpha) = \tan\alpha$.
- • A straight line through the origin with slope $\tan\alpha$ is precisely the radial line $\theta = \alpha$.
- • Therefore, the tangent line at the pole is simply $\theta = \alpha$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses polar angle with a horizontal line $y = \alpha$.
✗ **(C)** Confuses polar angle with a vertical line $x = \alpha$.
✗ **(D)** Assumes the slope is $f'(\alpha)$ rather than $\tan\alpha$.

Question 67 Analysis

★★★★ MANDATORY

Correct Answer: (C) $x + y = 2\sqrt{2}$

STEP-BY-STEP SOLUTION

- **Identify the Point in Rectangular Coordinates:**
- • $x = r\cos\theta = 2\cos(\pi/4) = 2(\sqrt{2}/2) = \sqrt{2}$.
- • $y = r\sin\theta = 2\sin(\pi/4) = 2(\sqrt{2}/2) = \sqrt{2}$.
- • Point of tangency is $(\sqrt{2}, \sqrt{2})$.
- **Find Tangent Slope:**
- • For any circle centered at the origin, the tangent line is perpendicular to the radial vector.
- • The radial line has slope $\tan(\pi/4) = 1$, so the tangent line has slope $m = -1$.
- **Write Equation of Tangent Line:**
- • $y - \sqrt{2} = -1(x - \sqrt{2}) \Rightarrow y - \sqrt{2} = -x + \sqrt{2} \Rightarrow x + y = 2\sqrt{2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the square root factor: $x + y = 2$.

✗ **(B)** Reports the normal (radial) line $y = x$ through the point instead of the tangent line.

✗ **(D)** Sign error in slope: $y = x + \sqrt{2}$.

Question 68 Analysis

★★★ IMPORTANT

Correct Answer: (D) $-\pi^2$

STEP-BY-STEP SOLUTION

- **Express y in terms of θ:**
- • $y = r\sin\theta = \theta^2\sin\theta$.
- **Differentiate using Product Rule:**
- • $dy/d\theta = d/d\theta\,[\theta^2\sin\theta] = 2\theta\sin\theta + \theta^2\cos\theta$.
- **Evaluate at $\theta = \pi$:**
- • $\sin(\pi) = 0$ and $\cos(\pi) = -1$.
- • $dy/d\theta = 2(\pi)(0) + (\pi)^2(-1) = 0 - \pi^2 = -\pi^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error in $\cos(\pi)$: $+\pi^2$.

✗ **(B)** Differentiates only the θ^2 factor: 2π.

✗ **(C)** Believes derivative is 0 because $\sin(\pi) = 0$, omitting the second product term.

Question 69 Analysis

★★★★ MANDATORY

Correct Answer: (A) -3

STEP-BY-STEP SOLUTION

- **Recall Polar Derivative Formula:**
- • $dy/dx = [(dr/d\theta)\sin\theta + r\cos\theta] / [(dr/d\theta)\cos\theta - r\sin\theta]$.
- **Evaluate at $\theta = \pi/4$:**
- • $\sin(\pi/4) = \sqrt{2}/2$ and $\cos(\pi/4) = \sqrt{2}/2$.
- • From the table: $r = 4$ and $dr/d\theta = 2$.
- • $dy/d\theta = (2)(\sqrt{2}/2) + 4(\sqrt{2}/2) = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$.
- • $dx/d\theta = (2)(\sqrt{2}/2) - 4(\sqrt{2}/2) = \sqrt{2} - 2\sqrt{2} = -\sqrt{2}$.
- **Compute dy/dx:**
- • $dy/dx = (3\sqrt{2}) / (-\sqrt{2}) = -3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error in $dx/d\theta$: $+3$.

✗ **(C)** Inverts the slope: $-1/3$.

✗ **(D)** Computes $(dr/d\theta)/2 = 1$.

Question 70 Analysis

★★★ IMPORTANT

Correct Answer: (B) θ = π / 3 and θ = 5π / 3

STEP-BY-STEP SOLUTION

- **Condition for Passing Through the Pole:**
- • A polar curve passes through the origin when r = 0.
- **Solve r = 0:**
- • 1 – 2 cos θ = 0 ⟹ 2 cos θ = 1 ⟹ cos θ = 1 / 2.
- **Find Angles in [0, 2π):**
- • In Quadrant I: θ = π / 3.
- • In Quadrant IV: θ = 2π – π/3 = 5π / 3.
- • Thus the curve passes through the pole at θ = π/3 and θ = 5π/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Solves sin θ = 1/2 instead of cos θ = 1/2: π/6 and 11π/6.
✗ **(C)** Solves cos θ = –1/2: 2π/3 and 4π/3.
✗ **(D)** Tests θ = 0 where r = 1 – 2 = –1 ≠ 0.

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2π

STEP-BY-STEP SOLUTION

- **Identify Limits for One Petal:**
- • The petal centered on the positive x-axis starts and ends where r = 0.
- • 4 cos(2θ) = 0 ⟹ 2θ = –π/2 and 2θ = π/2 ⟹ θ = –π/4 to θ = π/4.
- **Recall Polar Area Formula:** Area = (1/2) ∫_α^β r^2 dθ.
- **Set Up the Area Integral:**
- • Area = (1/2) ∫_(–π/4)^(π/4) [4 cos(2θ)]^2 dθ = (1/2) · 16 ∫_(–π/4)^(π/4) $\cos^2(2\theta)$ dθ.
- • Using symmetry across the x-axis: Area = 8 · 2 $\int_0^{\pi/4} \cos^2(2\theta)\, d\theta$ = 16 $\int_0^{\pi/4} \cos^2(2\theta)\, d\theta$.
- **Use Half-Angle Identity:** $\cos^2(2\theta) = (1 + \cos(4\theta)) / 2$.
- • Area = 16 $\int_0^{\pi/4} (1 + \cos(4\theta)) / 2\, d\theta$ = 8 [θ + (1/4) sin(4θ)]_0^(π/4).
- • At θ = π/4: 8 [π/4 + (1/4) sin(π)] = 8 [π/4 + 0] = 2π.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Halves the petal or integrates over [0, π/4] without doubling: π.
✗ **(B)** Forgets the 1/2 factor in the polar area formula: 4π.
✗ **(D)** Calculates the total area of all 4 petals: 4 · 2π = 8π.

Question 72 Analysis

★★★★ MANDATORY

Correct Answer: (D) 9π

STEP-BY-STEP SOLUTION

- **Determine Integration Limits:**
- • The circle r = 6 sin θ is completely traced as θ ranges from 0 to π.
- **Set Up Polar Area Integral:**
- • Area = (1/2) $\int_0^{\pi} (6 \sin\theta)^2\, d\theta$ = (1/2) · 36 $\int_0^{\pi} \sin^2\theta\, d\theta$ = 18 $\int_0^{\pi} \sin^2\theta\, d\theta$.
- **Evaluate the Integral:**
- • Recall $\int_0^{\pi} \sin^2\theta\, d\theta = \pi / 2$.
- • Area = 18 · (π / 2) = 9π.
- **Geometric Check:**
- • r = 6 sin θ is a circle of diameter 6, radius R = 3.
- • Area = π R^2 = π $(3)^2$ = 9π.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Halves the correct area: 9π / 2.
✗ **(B)** Integrates over [0, 2π], traversing the circle twice: 18π.
✗ **(C)** Uses diameter 6 as radius: π$(6)^2$ = 36π.

Question 73 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 27π / 2

STEP-BY-STEP SOLUTION

- **Limits of Integration:** The full cardioid is traced for $0 \le \theta \le 2\pi$.
- **Set Up Area Integral:**
- • Area = (1/2) $\int_{0}^{2\pi} (3 + 3\sin\theta)^2 \, d\theta$ = (1/2) $\int_{0}^{2\pi} (9 + 18\sin\theta + 9\sin^2\theta) \, d\theta$.
- **Integrate Term-by-Term:**
- • 1. $\int_{0}^{2\pi} 9 \, d\theta = 9(2\pi) = 18\pi$.
- • 2. $\int_{0}^{2\pi} 18\sin\theta \, d\theta = 0$ (integral of sine over full period).
- • 3. $\int_{0}^{2\pi} 9\sin^2\theta \, d\theta = 9 \cdot \pi = 9\pi$.
- **Combine Terms:**
- • Sum of integrals = $18\pi + 0 + 9\pi = 27\pi$.
- • Multiply by 1/2: Area = $27\pi / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the 1/2 factor in the polar area formula: 27π.
✗ **(C)** Omits the constant term or computes only the $\sin^2\theta$ contribution: 9π.
✗ **(D)** Evaluates only the constant term: $(1/2)(18\pi) = 9\pi$ or calculation mistake yielding 18π.

Question 74 Analysis

★★★★ MANDATORY

Correct Answer: (B) π / 3

STEP-BY-STEP SOLUTION

- **Find Boundaries of One Petal:**
- • Set $r = 0 \Rightarrow 2\sin(3\theta) = 0 \Rightarrow 3\theta = 0$ and $3\theta = \pi \Rightarrow \theta = 0$ to $\theta = \pi / 3$.
- **Set Up the Area Integral:**
- • Area = (1/2) $\int_{0}^{\pi/3} [2\sin(3\theta)]^2 \, d\theta$ = (1/2) · 4 $\int_{0}^{\pi/3} \sin^2(3\theta) \, d\theta$ = 2 $\int_{0}^{\pi/3} \sin^2(3\theta) \, d\theta$.
- **Evaluate the Definite Integral:**
- • Let $u = 3\theta \Rightarrow du = 3\,d\theta \Rightarrow d\theta = du / 3$.
- • When $\theta = 0$: $u = 0$; when $\theta = \pi/3$: $u = \pi$.
- • Area = 2 · (1/3) $\int_{0}^{\pi} \sin^2 u \, du$ = (2/3) · (π / 2) = π / 3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/2 factor in the polar area formula: $2\pi / 3$.
✗ **(C)** Halves the petal area: $\pi / 6$.
✗ **(D)** Computes the total area of all 3 petals: $3 \cdot (\pi/3) = \pi$.

Question 75 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) π – (3√3 / 2)

STEP-BY-STEP SOLUTION

- **Find Limits for the Inner Loop:**
- • The inner loop is formed where $r \le 0$, starting and ending at the pole $r = 0$.
- • $1 - 2\sin\theta = 0 \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi / 6$ and $\theta = 5\pi / 6$.
- **Set Up Area Integral:**
- • Area = (1/2) $\int_{\pi/6}^{5\pi/6} (1 - 2\sin\theta)^2 \, d\theta$ = (1/2) $\int_{\pi/6}^{5\pi/6} (1 - 4\sin\theta + 4\sin^2\theta) \, d\theta$.
- **Apply Half-Angle Identity:** $4\sin^2\theta = 2(1 - \cos(2\theta)) = 2 - 2\cos(2\theta)$.
- • Integrand = $(1 + 2) - 4\sin\theta - 2\cos(2\theta) = 3 - 4\sin\theta - 2\cos(2\theta)$.
- **Evaluate Antiderivative:**
- • $[3\theta + 4\cos\theta - \sin(2\theta)]$_(π/6)^(5π/6).
- • At $\theta = 5\pi/6$: $3(5\pi/6) + 4\cos(5\pi/6) - \sin(5\pi/3) = 5\pi/2 + 4(-\sqrt{3}/2) - (-\sqrt{3}/2) = 5\pi/2 - 2\sqrt{3} + \sqrt{3}/2 = 5\pi/2 - 3\sqrt{3}/2$.
- • At $\theta = \pi/6$: $3(\pi/6) + 4\cos(\pi/6) - \sin(\pi/3) = \pi/2 + 4(\sqrt{3}/2) - (\sqrt{3}/2) = \pi/2 + 2\sqrt{3} - \sqrt{3}/2 = \pi/2 + 3\sqrt{3}/2$.
- • Difference = $(5\pi/2 - \pi/2) - (3\sqrt{3}/2 + 3\sqrt{3}/2) = 2\pi - 3\sqrt{3}$.
- **Multiply by 1/2:**
- • Area = $(1/2)(2\pi - 3\sqrt{3}) = \pi - (3\sqrt{3} / 2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Halves the correct result: $\pi/2 - 3\sqrt{3}/4$.
✗ **(B)** Forgets to multiply by 1/2 from the polar area formula: $2\pi - 3\sqrt{3}$.
✗ **(D)** Sign error in integrating sine term: $\pi + 3\sqrt{3}/2$.

Question 76 Analysis

★★★ IMPORTANT

Correct Answer: (D) (1 / 2) $\int_0^{\pi/2} e^{4\theta} d\theta$

STEP-BY-STEP SOLUTION

- **Polar Area Formula:** Area = (1/2) $\int_\alpha^\beta r^2 \, d\theta$.
- **Square the Radius Function:**
- • $r = e^{2\theta} \Rightarrow r^2 = (e^{2\theta})^2 = e^{4\theta}$.
- **Substitute Limits:**
- • Area = (1/2) $\int_0^{\pi/2} e^{4\theta} \, d\theta$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits both the 1/2 factor and the squaring of r.
✗ **(B)** Forgets the 1/2 coefficient in front of the integral.
✗ **(C)** Forgets to square r, integrating $e^{2\theta}$ instead of $e^{4\theta}$.

Question 77 Analysis

★★★★ MANDATORY

Correct Answer: (A) 9

STEP-BY-STEP SOLUTION

- **Identify Loop Symmetry:**
- • One half-loop is traced in Quadrant I from $\theta = 0$ to $\theta = \pi / 4$ (where $\cos(2\theta) = 0$).
- • By symmetry, the total area of the two full loops is 4 times the area of this half-loop.
- **Set Up the Area Integral:**
- • Area = $4 \cdot [(1/2) \int_0^{\pi/4} r^2 \, d\theta] = 2 \int_0^{\pi/4} 9\cos(2\theta) \, d\theta$.
- • Notice $r^2 = 9\cos(2\theta)$ is already squared!
- **Evaluate the Definite Integral:**
- • $18 [(1/2) \sin(2\theta)]_0^{(\pi/4)} = 9 [\sin(2(\pi/4)) - \sin(0)] = 9 [\sin(\pi/2) - 0] = 9(1) = 9$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates $2 \cdot 9 = 18$ forgetting the 1/2 factor from antiderivative of $\sin(2\theta)$.
✗ **(C)** Calculates area of only one loop: 9 / 2.
✗ **(D)** Incorrectly inserts a factor of π, confusing lemniscate area with a circle.

Question 78 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $\int_0^k (2 + 2\cos\theta)^2 d\theta = 3\pi$

STEP-BY-STEP SOLUTION

- **Compute Total Area in Upper Half-Plane:**
- • Total Upper Area = (1/2) $\int_0^{\pi} (2 + 2\cos\theta)^2 \, d\theta$.
- • (1/2) $\int_0^{\pi} (4 + 8\cos\theta + 4\cos^2\theta) \, d\theta = (1/2) [4\pi + 0 + 4(\pi/2)] = (1/2) [4\pi + 2\pi] = 3\pi$.
- **Condition for Ray $\theta = k$ to Halve the Area:**
- • The area from $\theta = 0$ to $\theta = k$ must equal half of the upper area: $(1/2) \cdot 3\pi = 3\pi / 2$.
- • Therefore: (1/2) $\int_0^k (2 + 2\cos\theta)^2 \, d\theta = 3\pi / 2$.
- **Multiply Both Sides by 2:**
- • $\int_0^k (2 + 2\cos\theta)^2 \, d\theta = 3\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Equates to total area in both half-planes (6π).
✗ **(C)** Omits the squaring of the radius function.
✗ **(D)** Equates the integral without multiplying by 2.

Question 79 Analysis

★★★★ MANDATORY

Correct Answer: (C) $3\pi / 2$

STEP-BY-STEP SOLUTION

- **Recall the Polar Area Formula:** Area = $(1/2)\int^{0}^{\pi/2} [r(\theta)]^2 \, d\theta$.
- **Compute $[r(\theta)]^2$ at Each Given Angle:**
- • At $\theta = 0$: $[r(0)]^2 = 2^2 = 4$.
- • At $\theta = \pi/6$: $[r(\pi/6)]^2 = (\sqrt{7})^2 = 7$.
- • At $\theta = \pi/3$: $[r(\pi/3)]^2 = (\sqrt{7})^2 = 7$.
- • At $\theta = \pi/2$: $[r(\pi/2)]^2 = 2^2 = 4$.
- **Apply Trapezoidal Rule with $\Delta\theta = \pi / 6$:**
- • $\int^{0}^{\pi/2} [r(\theta)]^2 \, d\theta \approx (\Delta\theta / 2) \cdot [\, 4 + 2(7) + 2(7) + 4 \,]$.
- • $= ((\pi/6) / 2) \cdot [\, 4 + 14 + 14 + 4 \,] = (\pi / 12) \cdot 36 = 3\pi$.
- **Multiply by the Polar Constant (1/2):**
- • Area $\approx (1/2) \cdot (3\pi) = 3\pi / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by an extra factor of 2, obtaining $3\pi / 4$.

✗ **(B)** Forgets the 1/2 factor in front of the polar area integral, yielding 3π.

Question 80 Analysis

★★★ IMPORTANT

Correct Answer: (D) 4π

STEP-BY-STEP SOLUTION

- **Formula for Circular Sector Area:** Area = $(1/2) r^2 \Delta\theta$.
- **Compute the Angle Sweep $\Delta\theta$:**
- • $\Delta\theta = 2\pi / 3 - \pi / 6 = 4\pi / 6 - \pi / 6 = 3\pi / 6 = \pi / 2$.
- **Evaluate Sector Area:**
- • Area = $(1/2) \cdot (4)^2 \cdot (\pi / 2) = (1/2) \cdot 16 \cdot (\pi / 2) = 4\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates total area of the circle $\pi(4)^2 = 16\pi$.

✗ **(B)** Halves the correct sector area: 2π.

✗ **(C)** Doubles the correct area: 8π.

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $(1 / 2) \int^{\pi/3}^{5\pi/3} [\, 9 - (2 + 2\cos\theta)^2] \, d\theta$

STEP-BY-STEP SOLUTION

- **Find the Intersection Angles:**
- • Set the radii equal: $3 = 2 + 2\cos\theta \Longrightarrow 2\cos\theta = 1 \Longrightarrow \cos\theta = 1 / 2$.
- • On $[0, 2\pi]$, the intersection angles are $\theta = \pi / 3$ and $\theta = 5\pi / 3$.
- **Determine Which Curve is Outer:**
- • Test $\theta = \pi$: Circle $r = 3$; Cardioid $r = 2 + 2\cos(\pi) = 0$.
- • Thus, for $\pi / 3 \le \theta \le 5\pi / 3$, the circle is the outer boundary ($R = 3$) and the cardioid is the inner boundary ($r = 2 + 2\cos\theta$).
- **Set Up the Polar Area Integral:**
- • Area = $(1/2) \int^{\pi/3}^{5\pi/3} [\, R(\theta)^2 - r(\theta)^2] \, d\theta = (1 / 2) \int^{\pi/3}^{5\pi/3} [\, 9 - (2 + 2\cos\theta)^2] \, d\theta$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Integrates over $[-\pi/3, \pi/3]$, which is the region where the cardioid lies outside the circle.

✗ **(C)** Subtracts the radius functions without squaring them.

✗ **(D)** Reverses the inner and outer curves, resulting in a negative area.

Question 82 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) π / 2 – 1

STEP-BY-STEP SOLUTION

- **Find the Point of Intersection:**
- • In Quadrant I, $2 \sin\theta = 2 \cos\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \pi / 4$.
- **Analyze the Boundary Behavior:**
- • For $0 \le \theta \le \pi / 4$, the curve closer to the pole is $r = 2 \sin\theta$ (starts at 0).
- • For $\pi / 4 \le \theta \le \pi / 2$, the curve closer to the pole is $r = 2 \cos\theta$ (drops to 0).
- **Set Up and Evaluate by Symmetry:**
- • Area = $(1/2) \int_{0}^{\pi/4} (2 \sin\theta)^2 \, d\theta + (1/2) \int_{\pi/4}^{\pi/2} (2 \cos\theta)^2 \, d\theta$.
- • By symmetry across $\theta = \pi / 4$, Area = $2 \cdot [(1/2) \int_{0}^{\pi/4} 4 \sin^2\theta \, d\theta] = 4 \int_{0}^{\pi/4} \sin^2\theta \, d\theta$.
- • Using $\sin^2\theta = (1 - \cos(2\theta)) / 2$:
- • $4 \cdot [\theta / 2 - \sin(2\theta) / 4]_0^{(\pi/4)} = 4 [(\pi / 8 - 1 / 4) - 0] = \pi / 2 - 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by 2 when computing the single symmetrical half.

✗ **(C)** Approximates the region as a triangle or uses an incorrect antiderivative.

✗ **(D)** Calculates the area of half of one circle: $(1/2)\pi(1)^2 = \pi / 2$.

Question 83 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2π / 3 + √3

STEP-BY-STEP SOLUTION

- **Find the Intersection Angles:**
- • $4 \cos(2\theta) = 2 \Rightarrow \cos(2\theta) = 1 / 2 \Rightarrow 2\theta = \pm\pi / 3 \Rightarrow \theta = \pm\pi / 6$.
- **Set Up the Area Integral:**
- • Area = $(1/2) \int_{-\pi/6}^{\pi/6} [(4 \cos(2\theta))^2 - 2^2] \, d\theta$.
- • By symmetry: Area = $\int_{0}^{\pi/6} [16 \cos^2(2\theta) - 4] \, d\theta$.
- **Simplify the Integrand:**
- • $16 \cos^2(2\theta) - 4 = 16 [(1 + \cos(4\theta)) / 2] - 4 = 8 + 8 \cos(4\theta) - 4 = 4 + 8 \cos(4\theta)$.
- **Integrate:**
- • $[4\theta + 2 \sin(4\theta)]_0^{(\pi/6)} = 4(\pi / 6) + 2 \sin(2\pi / 3) = 2\pi / 3 + 2(\sqrt{3} / 2) = 2\pi / 3 + \sqrt{3}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error in integrating $\cos(4\theta)$, getting $-\sqrt{3}$ instead of $+\sqrt{3}$.

✗ **(B)** Forgets the 1/2 factor before applying symmetry: $4\pi / 3 + 2\sqrt{3}$.

✗ **(D)** Omits the trigonometric oscillation term $2 \sin(2\pi / 3)$.

Question 84 Analysis

★★★★ MANDATORY

Correct Answer: (D) 19 / 3

STEP-BY-STEP SOLUTION

- **Recall the Polar Arc Length Formula:**
- • $L = \int_{\alpha}^{\beta} \sqrt{[r^2 + (dr/d\theta)^2]} \, d\theta$.
- **Differentiate and Form the Integrand:**
- • $r = \theta^2 \Rightarrow dr/d\theta = 2\theta$.
- • $r^2 + (dr/d\theta)^2 = (\theta^2)^2 + (2\theta)^2 = \theta^4 + 4\theta^2 = \theta^2(\theta^2 + 4)$.
- • Since $\theta \ge 0$: $\sqrt{[\theta^2(\theta^2 + 4)]} = \theta \sqrt{(\theta^2 + 4)}$.
- **Evaluate the Definite Integral:**
- • $L = \int_{0}^{\sqrt{5}} \theta \sqrt{(\theta^2 + 4)} \, d\theta$.
- • Let $u = \theta^2 + 4 \Rightarrow du = 2\theta \, d\theta \Rightarrow \theta \, d\theta = du / 2$.
- • When $\theta = 0$, $u = 4$; when $\theta = \sqrt{5}$, $u = (\sqrt{5})^2 + 4 = 9$.
- • $L = (1/2) \int_{4}^{9} u^{(1/2)} \, du = (1/2) [(2/3) u^{(3/2)}]_4^9 = (1/3) [9^{(3/2)} - 4^{(3/2)}] = (1/3) [27 - 8] = 19 / 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/3 coefficient from integration.

✗ **(B)** Forgets the factor of 1/2 from du substitution, obtaining 38 / 3.

✗ **(C)** Evaluates only at lower limit or makes an arithmetic error.

Question 85 Analysis

★★★★ MANDATORY

Correct Answer: (A) (√10 / 3) (e⁶ – 1)

STEP-BY-STEP SOLUTION

- **Compute the Derivative:**
- • $r = e^{3\theta} \Rightarrow dr/d\theta = 3e^{3\theta}$.
- **Set Up the Polar Arc Length Radicand:**
- • $r^2 + (dr/d\theta)^2 = (e^{3\theta})^2 + (3e^{3\theta})^2 = e^{6\theta} + 9e^{6\theta} = 10e^{6\theta}$.
- • $\sqrt{[r^2 + (dr/d\theta)^2]} = \sqrt{(10e^{6\theta})} = \sqrt{10}\, e^{3\theta}$.
- **Integrate Over [0, 2]:**
- • $L = \int_0^2 \sqrt{10}\, e^{3\theta}\, d\theta = \sqrt{10}\, [(1/3)\, e^{3\theta}]_0^2 = (\sqrt{10}/3)\, [e^6 - e^0] = (\sqrt{10}/3)\, (e^6 - 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by 3 when integrating $e^{3\theta}$.
✗ **(C)** Misses the √10 factor from the metric $r^2 + (r')^2$.
✗ **(D)** Incorrectly divides by 6 instead of 3.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 6

STEP-BY-STEP SOLUTION

- **Relate Swept Area to Angle θ:**
- • The area swept out from 0 to θ is $A(\theta) = (1/2) \int^0 \wedge \theta\, [r(u)]^2\, du$.
- • By the Fundamental Theorem of Calculus: $dA/d\theta = (1/2)\, [r(\theta)]^2$.
- **Apply the Chain Rule to Find dA/dt:**
- • $dA/dt = (dA/d\theta) \cdot (d\theta/dt) = (1/2)\, [r(\theta)]^2 \cdot (d\theta/dt)$.
- **Substitute Given Values at θ = π / 3:**
- • $r(\pi/3) = 4\cos(\pi/3) = 4(1/2) = 2$.
- • $dA/dt = (1/2) \cdot (2)^2 \cdot 3 = (1/2) \cdot 4 \cdot 3 = 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/2 factor in the polar area formula: (4)(3) = 12.
✗ **(C)** Omits squaring r, computing (1/2)(2)(3) = 3.
✗ **(D)** Uses r = 4 cos(0) = 4 instead of evaluating at θ = π/3.

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 6

STEP-BY-STEP SOLUTION

- **Understand the Region's Boundary:**
- • The perimeter consists of two parts: (1) the curved arc of the cardioid, and (2) the straight segment along the x-axis.
- **Compute the Curved Arc Length:**
- • $r = 1 + \cos\theta \Rightarrow dr/d\theta = -\sin\theta$.
- • $r^2 + (dr/d\theta)^2 = (1 + \cos\theta)^2 + (-\sin\theta)^2 = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta$.
- • Using half-angle identity $2(1 + \cos\theta) = 4\cos^2(\theta/2)$:
- • $\sqrt{[r^2 + (dr/d\theta)^2]} = 2\cos(\theta/2)$ on $[0, \pi]$.
- • $L_curve = \int^0 \pi\, 2\cos(\theta/2)\, d\theta = [4\sin(\theta/2)]_0 \wedge \pi = 4\sin(\pi/2) - 0 = 4$.
- **Compute the Straight Boundary Length Along x-axis:**
- • At θ = 0: r = 1 + cos(0) = 2 (the point (2, 0)).
- • At θ = π: r = 1 + cos(π) = 0 (the pole (0, 0)).
- • The straight line segment along the x-axis connects (0, 0) to (2, 0), which has length 2.
- **Total Perimeter:**
- • Perimeter = L_curve + L_segment = 4 + 2 = 6.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the perimeter with circle circumference approximations.
✗ **(B)** Forgets the straight line segment along the x-axis, reporting only the arc length 4.
✗ **(D)** Arithmetic error combining the arc and base.

Question 88 Analysis

★★★★ MANDATORY

Correct Answer: (D) 19π / 4

STEP-BY-STEP SOLUTION

- **Recall Arc Length Formula:** $L = \int^{0}_{\pi/2} \sqrt{[(f(\theta))^2 + (f'(\theta))^2]}\, d\theta$.
- **Compute the Speed Metric $g(\theta) = \sqrt{[(f(\theta))^2 + (f'(\theta))^2]}$ at Each Point:**
- • At $\theta = 0$: $\sqrt{[3^2 + 4^2]} = \sqrt{25} = 5$.
- • At $\theta = \pi/4$: $\sqrt{[6^2 + 8^2]} = \sqrt{100} = 10$.
- • At $\theta = \pi/2$: $\sqrt{[5^2 + 12^2]} = \sqrt{169} = 13$.
- **Apply Trapezoidal Rule with $\Delta\theta = \pi / 4$:**
- • $L \approx (\Delta\theta / 2) \cdot [g(0) + 2\, g(\pi/4) + g(\pi/2)]$.
- • $= ((\pi/4) / 2) \cdot [5 + 2(10) + 13] = (\pi / 8) \cdot [5 + 20 + 13] = (\pi / 8) \cdot 38 = 19\pi / 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $\Delta\theta$ instead of $\Delta\theta / 2$ in the trapezoid formula.
✗ **(B)** Forgets to multiply the interior term $g(\pi/4)$ by 2: $(\pi/8)(5 + 10 + 13) = 28\pi/8$.
✗ **(C)** Omits the $(\Delta\theta / 2)$ factor completely.

Question 89 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) π – (3√3 / 2)

STEP-BY-STEP SOLUTION

- **Determine Where the Inner Loop Begins and Ends:**
- • The curve passes through the pole when $r = 0 \Rightarrow 1 + 2\cos\theta = 0 \Rightarrow \cos\theta = -1/2$.
- • On $[0, 2\pi]$, the roots are $\theta = 2\pi / 3$ and $\theta = 4\pi / 3$.
- • The inner loop is traced for $2\pi / 3 \le \theta \le 4\pi / 3$.
- **Set Up Area by Symmetry:**
- • By symmetry across $\theta = \pi$: Area $= 2 \cdot [(1/2) \int^{2\pi/3}_{\pi} (1 + 2\cos\theta)^2\, d\theta] = \int^{2\pi/3}_{\pi} (1 + 4\cos\theta + 4\cos^2\theta)\, d\theta$.
- **Rewrite with Double-Angle Identity:**
- • $4\cos^2\theta = 2(1 + \cos(2\theta)) = 2 + 2\cos(2\theta)$.
- • Integrand $= 1 + 4\cos\theta + 2 + 2\cos(2\theta) = 3 + 4\cos\theta + 2\cos(2\theta)$.
- **Evaluate the Definite Integral:**
- • $[3\theta + 4\sin\theta + \sin(2\theta)]_(2\pi/3)\text{^}\pi$.
- • At π: $3\pi + 4(0) + 0 = 3\pi$.
- • At $2\pi/3$: $3(2\pi/3) + 4\sin(2\pi/3) + \sin(4\pi/3) = 2\pi + 4(\sqrt{3} / 2) + (-\sqrt{3} / 2) = 2\pi + 3\sqrt{3} / 2$.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 4π / 3 + 2√3

STEP-BY-STEP SOLUTION

- **Find the Intersection Points:**
- • $4\sin\theta = 2 \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi / 6$ and $\theta = 5\pi / 6$.
- **Set Up the Area Integral:**
- • Area $= (1/2) \int^{5\pi/6}_{\pi/6} [(4\sin\theta)^2 - 2^2]\, d\theta$.
- • By symmetry across $\theta = \pi / 2$: Area $= \int^{\pi/2}_{\pi/6} [16\sin^2\theta - 4]\, d\theta$.
- **Simplify the Integrand:**
- • $16\sin^2\theta - 4 = 16[(1 - \cos(2\theta)) / 2] - 4 = 8 - 8\cos(2\theta) - 4 = 4 - 8\cos(2\theta)$.
- **Integrate:**
- • $[4\theta - 4\sin(2\theta)]_(\pi/6)\text{^}(\pi/2)$.
- • At $\pi/2$: $4(\pi / 2) - 4\sin(\pi) = 2\pi - 0 = 2\pi$.
- • At $\pi/6$: $4(\pi / 6) - 4\sin(\pi / 3) = 2\pi / 3 - 4(\sqrt{3} / 2) = 2\pi / 3 - 2\sqrt{3}$.
- • Area $= 2\pi - (2\pi / 3 - 2\sqrt{3}) = 4\pi / 3 + 2\sqrt{3}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only one half without multiplying by 2: $2\pi / 3 + \sqrt{3}$.
✗ **(C)** Sign error on the antiderivative of $-8\cos(2\theta)$.
✗ **(D)** Integrates without subtracting the inner circle $r = 2$.

FRQ 1: Parametric Motion & Velocity Table Analysis Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Tangent Slope & Tangent Line at t = 3 (2 Points)

- **Compute Parametric Slope dy/dx:**

$dy/dx = y'(t) / x'(t)$.

At $t = 3$: $dy/dx|_{t=3} = y'(3) / x'(3) = 2 / 5$.

- **Identify Point of Tangency:**

$(x(3), y(3)) = (4, 3)$.

- **Write Tangent Line Equation:**

$y - 3 = (2 / 5)(x - 4)$ (or $y = (2/5)x + 7/5$).

Part (b): Speed & Rate of Change of Speed at t = 0 (2 Points)

- **Compute Speed at t = 0:**

Speed $= ||v(0)|| = \sqrt{[(x'(0))^2 + (y'(0))^2]} = \sqrt{[2^2 + (-1)^2]} = \sqrt{[4 + 1]} = \sqrt{5}$.

- **Evaluate Derivative of Speed:**

$d/dt\,||v(t)|| = [x'(t)x''(t) + y'(t)y''(t)] / \sqrt{[(x'(t))^2 + (y'(t))^2]}$.

At $t = 0$: $d/dt\,||v(t)|| = [(2)(3) + (-1)(2)] / \sqrt{5} = [6 - 2] / \sqrt{5} = 4 / \sqrt{5}$.

- **Conclusion:**

Since $d/dt\,||v(t)|| = 4/\sqrt{5} > 0$, the speed of the particle is **increasing** at $t = 0$.

Part (c): Tangent Orientation at t = 6 (2 Points)

- **Evaluate Component Derivatives at t = 6:**

From the table: $x'(6) = 0$ and $y'(6) = 4$.

- **Analyze dy/dx:**

Since $dx/dt = 0$ while $dy/dt \neq 0$, the slope $dy/dx = 4 / 0$ is undefined.

- **Conclusion:**

The path of the particle has a **vertical tangent line** at $t = 6$.

Part (d): Trapezoidal Sum for Total Distance (3 Points)

- **Formula for Distance Traveled:**

Distance $= \int_0^{12} \sqrt{[(x'(t))^2 + (y'(t))^2]}\,dt$.

- **Calculate Speed s(t) at Each Table Grid Point:**

$s(0) = \sqrt{[2^2 + (-1)^2]} = \sqrt{5}$

$s(3) = \sqrt{[5^2 + 2^2]} = \sqrt{29}$

$s(6) = \sqrt{[0^2 + 4^2]} = 4$

$s(9) = \sqrt{[(-3)^2 + 0^2]} = 3$

$s(12) = \sqrt{[(-4)^2 + (-2)^2]} = \sqrt{20} = 2\sqrt{5}$

- **Set Up Trapezoidal Approximation with Δt = 3:**

Distance $\approx (\Delta t / 2) \cdot [s(0) + 2s(3) + 2s(6) + 2s(9) + s(12)]$

$= (3 / 2) \cdot [\sqrt{5} + 2\sqrt{29} + 2(4) + 2(3) + 2\sqrt{5}]$

$= (3 / 2) \cdot [3\sqrt{5} + 2\sqrt{29} + 14]$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Slope of tangent line: 1 pt Tangent line equation: 1 pt	**2**
Part (b)	Speed at t = 0: 1 pt Increasing conclusion with sign of d(speed)/dt: 1 pt	**2**
Part (c)	Vertical tangent identification: 1 pt Reasoning using $x'(6) = 0$ and $y'(6) \neq 0$: 1 pt	**2**
Part (d)	Speed values computed at each t: 1 pt Trapezoidal sum formula setup: 1 pt Final numerical approximation: 1 pt	**3**

FRQ 2: Planar Vector Motion & Particle Kinematics Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Position Vector via Definite Integration (3 Points)

- **Integrate Horizontal Velocity:**

$x(t) = x(2) + \int_2^t (3u^2 - 12)\,du = 5 + [u^3 - 12u]_2^t$

$= 5 + (t^3 - 12t) - (2^3 - 12(2)) = 5 + t^3 - 12t - (8 - 24) = 5 + t^3 - 12t + 16 = t^3 - 12t + 21.$

- **Integrate Vertical Velocity:**

$y(t) = y(2) + \int_2^t (4e^{2u-4} - 4)\,du = -3 + [2e^{2u-4} - 4u]_2^t$

$= -3 + (2e^{2t-4} - 4t) - (2e^0 - 4(2)) = -3 + 2e^{2t-4} - 4t - (2 - 8) = -3 + 2e^{2t-4} - 4t + 6 = 2e^{2t-4} - 4t + 3.$

- **Position Vector:**

$r(t) = \langle t^3 - 12t + 21,\ 2e^{2t-4} - 4t + 3 \rangle.$

Part (b): Times When the Particle is at Rest (2 Points)

- **Condition for Rest:**

A particle is at rest if and only if both velocity components are zero simultaneously: $v(t) = \langle 0, 0 \rangle$.

- **Horizontal Component:**

$3t^2 - 12 = 0 \Rightarrow 3(t^2 - 4) = 0 \Rightarrow t = 2$ (since $t > 0$).

- **Vertical Component:**

$4e^{2t-4} - 4 = 0 \Rightarrow e^{2t-4} = 1 \Rightarrow 2t - 4 = 0 \Rightarrow t = 2.$

- **Conclusion:**

Both components vanish at exactly **t = 2**.

Part (c): Acceleration Vector & Direction Angle (2 Points)

- **Differentiate Velocity:**

$a(t) = v'(t) = \langle d/dt(3t^2 - 12),\ d/dt(4e^{2t-4} - 4) \rangle = \langle 6t,\ 8e^{2t-4} \rangle.$

- **Evaluate at t = 2:**

$a(2) = \langle 6(2),\ 8e^0 \rangle = \langle 12, 8 \rangle.$

- **Direction Angle:**

Since both components are positive (Quadrant I):

$\tan\theta = 8 / 12 = 2 / 3 \Rightarrow \theta = \arctan(2/3) \approx 0.588$ radians (or 33.69°).

Part (d): Distance Traveled Integral (2 Points)

- **Speed Function:**

$||v(t)|| = \sqrt{[(3t^2 - 12)^2 + (4e^{2t-4} - 4)^2]}.$

- **Integral Expression:**

Total Distance $= \int_1^3 \sqrt{[(3t^2 - 12)^2 + (4e^{2t-4} - 4)^2]}\,dt.$

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Antiderivatives of velocity components: 1 pt Initial condition constants: 1 pt Final position vector: 1 pt	**3**
Part (b)	Sets $v(t) = \langle 0, 0 \rangle$: 1 pt Identifies $t = 2$ as unique positive solution: 1 pt	**2**
Part (c)	Acceleration vector $a(2) = \langle 12, 8 \rangle$: 1 pt Direction angle $\theta = \arctan(2/3)$: 1 pt	**2**
Part (d)	Integrand for speed: 1 pt Limits of integration [1, 3]: 1 pt	**2**

FRQ 3: Polar Area Bounded by Two Curves Official Model Solution ★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Polar Coordinates of Intersections (2 Points)

- **Set the Radii Equal:**

$2 + 2\cos\theta = 3 \Rightarrow 2\cos\theta = 1 \Rightarrow \cos\theta = 1/2$.

- **Solve for Angles on [0, 2π):**

$\theta = \pi/3$ and $\theta = 5\pi/3$.

- **Coordinates of Points:**

Since $r = 3$ at both angles, the points are **$(3, \pi/3)$** and **$(3, 5\pi/3)$**.

Part (b): Exact Area of Region R (3 Points)

- **Identify Inner and Outer Boundaries:**

For $\pi/3 \le \theta \le 5\pi/3$, test $\theta = \pi$: Circle $r = 3$, Cardioid $r = 0$.

Thus the circle is the outer boundary and the cardioid is the inner boundary.

- **Set Up Integral by Symmetry:**

$\text{Area}(R) = (1/2)\int_{\pi/3}^{5\pi/3} [3^2 - (2 + 2\cos\theta)^2]\, d\theta = \int_{\pi/3}^{\pi} [9 - (4 + 8\cos\theta + 4\cos^2\theta)]\, d\theta$.

- **Apply Double-Angle Identity $4\cos^2\theta = 2 + 2\cos(2\theta)$:**

Integrand $= 9 - 4 - 8\cos\theta - 2 - 2\cos(2\theta) = 3 - 8\cos\theta - 2\cos(2\theta)$.

- **Evaluate Antiderivative:**

$[3\theta - 8\sin\theta - \sin(2\theta)]_{\pi/3}^{\pi}$

At π: $3(\pi) - 8\sin(\pi) - \sin(2\pi) = 3\pi$.

At $\pi/3$: $3(\pi/3) - 8\sin(\pi/3) - \sin(2\pi/3) = \pi - 8(\sqrt{3}/2) - (\sqrt{3}/2) = \pi - 9\sqrt{3}/2$.

- **Subtract:**

$\text{Area}(R) = 3\pi - (\pi - 9\sqrt{3}/2) =$ **$2\pi + (9\sqrt{3}/2)$**.

Part (c): Integral Setup for Region S (2 Points)

- **Boundary Behavior for S:**

For $-\pi/3 \le \theta \le \pi/3$, the cardioid lies outside the circle: $R_{outer} = 2 + 2\cos\theta$, $r_{inner} = 3$.

- **Area Integral:**

$\text{Area}(S) = (1/2)\int_{-\pi/3}^{\pi/3} [(2 + 2\cos\theta)^2 - 9]\, d\theta$ (or $\int_{0}^{\pi/3} [(2 + 2\cos\theta)^2 - 9]\, d\theta$).

Part (d): Area Bisection Ray (2 Points)

- **Value of k:** $k = \pi$.
- **Symmetry Justification:**

The cardioid $r = 2 + 2\cos\theta$ is symmetric with respect to the polar axis (the line containing $\theta = 0$ and $\theta = \pi$).

Therefore, the ray **$\theta = \pi$** divides the cardioid into two congruent halves of identical area.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Equates radii: 1 pt Intersection coordinates $(3, \pi/3)$ and $(3, 5\pi/3)$: 1 pt	**2**
Part (b)	Integral setup with correct limits: 1 pt Antidifferentiation: 1 pt Exact area $2\pi + (9\sqrt{3}/2)$: 1 pt	**3**
Part (c)	Integrand $[(2 + 2\cos\theta)^2 - 9]$: 1 pt Correct limits and factor of 1/2: 1 pt	**2**
Part (d)	$k = \pi$: 1 pt Justification via symmetry across polar axis: 1 pt	**2**

FRQ 4: Polar Slopes & Tangent Lines for a Limacon Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Derivation of dy/dx & Slope at $\theta = \pi / 6$ (3 Points)

- **Express y in Terms of θ:**

$y = r \sin\theta = (2 - 2\sin\theta)\sin\theta = 2\sin\theta - 2\sin^2\theta$.

$dy/d\theta = 2\cos\theta - 4\sin\theta\cos\theta = 2\cos\theta(1 - 2\sin\theta)$.

- **Express x in Terms of θ:**

$x = r\cos\theta = (2 - 2\sin\theta)\cos\theta = 2\cos\theta - 2\sin\theta\cos\theta = 2\cos\theta - \sin(2\theta)$.

$dx/d\theta = -2\sin\theta - 2\cos(2\theta)$.

- **Evaluate at $\theta = \pi / 6$:**

$\sin(\pi/6) = 1/2$, $\cos(\pi/6) = \sqrt{3}/2$, $\cos(2(\pi/6)) = \cos(\pi/3) = 1/2$.

$dy/d\theta = 2(\sqrt{3}/2)[1 - 2(1/2)] = \sqrt{3}[1 - 1] = 0$.

$dx/d\theta = -2(1/2) - 2(1/2) = -1 - 1 = -2 \neq 0$.

- **Slope:**

$dy/dx = (dy/d\theta) / (dx/d\theta) = 0 / (-2) = \mathbf{0}$.

Part (b): Coordinates & Tangent Line Equation at $\theta = \pi / 6$ (2 Points)

- **Find r and Cartesian Coordinates:**

$r(\pi/6) = 2 - 2(1/2) = 1$.

$x = r\cos(\pi/6) = 1 \cdot (\sqrt{3}/2) = \sqrt{3}/2$.

$y = r\sin(\pi/6) = 1 \cdot (1/2) = 1/2$.

- **Write Tangent Line:**

With slope $m = 0$ through $(\sqrt{3}/2, 1/2)$: $\mathbf{y = 1/2}$.

Part (c): All Angles with Horizontal Tangents (2 Points)

- **Condition for Horizontal Tangent:**

$dy/d\theta = 0$ and $dx/d\theta \neq 0$.

- **Solve $dy/d\theta = 2\cos\theta(1 - 2\sin\theta) = 0$:**

1. $\cos\theta = 0 \Rightarrow \theta = \pi/2$ or $\theta = 3\pi/2$.

• At $\theta = \pi/2$: $dx/d\theta = -2(1) - 2\cos(\pi) = -2 + 2 = 0$ (pole, tangent is vertical).

• At $\theta = 3\pi/2$: $dx/d\theta = -2(-1) - 2\cos(3\pi) = 2 - 2(-1) = 4 \neq 0 \Rightarrow$ Horizontal!

2. $1 - 2\sin\theta = 0 \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi/6$ or $\theta = 5\pi/6$.

• At $\theta = 5\pi/6$: $dx/d\theta = -2(1/2) - 2\cos(5\pi/3) = -1 - 2(1/2) = -2 \neq 0 \Rightarrow$ Horizontal!

- **Solutions:**

The horizontal tangents occur at $\mathbf{\theta = \pi/6,\ 5\pi/6,\ \text{and}\ 3\pi/2}$.

Part (d): Tangent at the Pole (2 Points)

- **Pole Angle:**

$r = 0 \Rightarrow 2 - 2\sin\theta = 0 \Rightarrow \sin\theta = 1 \Rightarrow \theta = \pi/2$.

- **Tangent Line at the Pole:**

For any polar curve passing through the pole at $\theta = \theta_0$, the tangent line is the line $\theta = \theta_0$.

Here, the tangent line is the ray $\theta = \pi/2$ (the y-axis), whose equation is $\mathbf{x = 0}$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Derivative expressions: 1 pt Evaluation at $\theta = \pi/6$: 1 pt Slope $dy/dx = 0$: 1 pt	**3**
Part (b)	Cartesian coordinates $(\sqrt{3}/2, 1/2)$: 1 pt Horizontal tangent equation $y = 1/2$: 1 pt	**2**
Part (c)	Sets $dy/d\theta = 0$: 1 pt Verifies $dx/d\theta \neq 0$ giving $\theta = \pi/6, 5\pi/6, 3\pi/2$: 1 pt	**2**
Part (d)	Pole angle $\theta = \pi/2$: 1 pt Tangent line at pole is $x = 0$ (vertical line): 1 pt	**2**

FRQ 5: Applied Drone Flight & Parametric Arc Length Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Drone Position at t = 4 (2 Points)

- **Integrate x'(t):**

$x(4) = x(0) + \int_0^4 6t^{(1/2)}\,dt = 0 + [\,4t^{(3/2)}\,]_0^4 = 4(4^{(3/2)}) = 4(8) = 32.$

- **Integrate y'(t):**

$y(4) = y(0) + \int_0^4 (3t - 3)\,dt = 4 + [\,(3/2)t^2 - 3t\,]_0^4 = 4 + [\,(3/2)(16) - 12\,] = 4 + [\,24 - 12\,] = 16.$

- **Position:** (32, 16).

Part (b): Speed at t = 4 (2 Points)

- **Compute Component Velocities:**

$x'(4) = 6\sqrt{4} = 12.$

$y'(4) = 3(4) - 3 = 12 - 3 = 9.$

- **Compute Speed:**

$||v(4)|| = \sqrt{[\,(x'(4))^2 + (y'(4))^2\,]} = \sqrt{[\,12^2 + 9^2\,]} = \sqrt{[\,144 + 81\,]} = \sqrt{225} = \mathbf{15}.$

Part (c): Exact Total Distance from t = 0 to t = 4 (3 Points)

- **Form the Speed Integrand:**

$(x'(t))^2 + (y'(t))^2 = (6\sqrt{t})^2 + (3t - 3)^2 = 36t + (9t^2 - 18t + 9) = 9t^2 + 18t + 9 = 9(t^2 + 2t + 1) = 9(t + 1)^2.$

- **Take the Square Root:**

Since $t \geq 0$: $\sqrt{[\,9(t + 1)^2\,]} = 3(t + 1) = 3t + 3.$

- **Integrate Over [0, 4]:**

Distance = $\int_0^4 (3t + 3)\,dt = [\,(3/2)t^2 + 3t\,]_0^4 = (3/2)(16) + 3(4) = 24 + 12 = \mathbf{36}.$

Part (d): Distance Integral for [4, 10] (2 Points)

- **Set Up Integral:**

Distance = $\int_4^{10} 3(t + 1)\,dt$ (or $\int_4^{10} \sqrt{[\,36t + (3t - 3)^2\,]}\,dt$).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	x(4) = 32: 1 pt y(4) = 16: 1 pt	2
Part (b)	Speed formula setup: 1 pt Speed = 15: 1 pt	2
Part (c)	Simplification of speed radicand to 3(t + 1): 1 pt Antiderivative: 1 pt Total distance = 36: 1 pt	3
Part (d)	Integral expression over [4, 10]: 2 pts	2

FRQ 6: Vector Acceleration & Maximum Height Flight Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Velocity and Position Vectors (3 Points)

- **Integrate Acceleration:**

$v(t) = v(0) + \int_0^t \langle -2, 6 - 6u \rangle \, du = \langle 10 - 2t, 9 + 6t - 3t^2 \rangle$.

- **Integrate Velocity to Obtain Position:**

$r(t) = r(0) + \int_0^t \langle 10 - 2u, 9 + 6u - 3u^2 \rangle \, du = \langle 10t - t^2, 9t + 3t^2 - t^3 \rangle$.

Part (b): Maximum Vertical Height (2 Points)

- **Set Vertical Velocity y'(t) = 0:**

$9 + 6t - 3t^2 = 0 \Rightarrow -3(t^2 - 2t - 3) = 0 \Rightarrow -3(t - 3)(t + 1) = 0$.

Since $t > 0$, $t = 3$.

- **Confirm Maximum:**

$y''(3) = a_y(3) = 6 - 6(3) = -12 < 0$ (concave down $\Rightarrow$ local/absolute maximum).

- **Compute Height:**

$y(3) = 9(3) + 3(3)^2 - 3^3 = 27 + 27 - 27 = \mathbf{27}$.

Part (c): Speed & Acceleration Magnitude at t = 3 (2 Points)

- **Evaluate Velocity Vector at t = 3:**

$v(3) = \langle 10 - 2(3), 0 \rangle = \langle 4, 0 \rangle$.

Speed $= ||v(3)|| = \sqrt{[4^2 + 0^2]} = \mathbf{4}$.

- **Evaluate Acceleration Vector at t = 3:**

$a(3) = \langle -2, 6 - 6(3) \rangle = \langle -2, -12 \rangle$.

Magnitude $= ||a(3)|| = \sqrt{[(-2)^2 + (-12)^2]} = \sqrt{[4 + 144]} = \sqrt{148} = \mathbf{2\sqrt{37}}$.

Part (d): Behavior of Speed at Maximum Height (2 Points)

- **Compute the Dot Product v(3) · a(3):**

$v(3) \cdot a(3) = (4)(-2) + (0)(-12) = -8 + 0 = -8$.

- **Relate Dot Product to Derivative of Speed:**

$d/dt \, ||v(t)|| = [v(t) \cdot a(t)] / ||v(t)|| = -8 / 4 = -2$.

- **Conclusion:**

Because the derivative of speed is negative ($v \cdot a < 0$), the speed of the projectile is **decreasing** at $t = 3$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Velocity vector $v(t) = \langle 10 - 2t, 9 + 6t - 3t^2 \rangle$: 1 pt Position vector $r(t) = \langle 10t - t^2, 9t + 3t^2 - t^3 \rangle$: 2 pts	3
Part (b)	Critical time $t = 3$: 1 pt Maximum vertical height $y(3) = 27$: 1 pt	2
Part (c)	Speed = 4: 1 pt Acceleration magnitude = $2\sqrt{37}$: 1 pt	2
Part (d)	Evaluation of $v(3) \cdot a(3) = -8 < 0$: 1 pt Conclusion that speed is decreasing: 1 pt	2

FRQ 7: Polar Rose Curve Petals, Perimeter & Radial Rates Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Quadrant I Petal Interval and Enclosed Area (3 Points)

- **Determine Interval for First Petal:**

The curve leaves the pole when $r = 0 \Rightarrow 3\sin(2\theta) = 0 \Rightarrow 2\theta = 0 \Rightarrow \theta = 0$.

The curve returns to the pole when $2\theta = \pi \Rightarrow \theta = \pi / 2$.

Thus the Quadrant I petal is traced on $\mathbf{0 \le \theta \le \pi / 2}$.

- **Set Up and Evaluate the Area Integral:**

Area = $(1/2) \int_{0}^{\pi/2} [3\sin(2\theta)]^2 \, d\theta = (9/2) \int_{0}^{\pi/2} \sin^2(2\theta) \, d\theta$.

- **Apply Power-Reduction Identity $\sin^2(2\theta) = (1 - \cos(4\theta)) / 2$:**

$= (9 / 4) [\theta - \sin(4\theta) / 4]_0^{(\pi/2)} = (9 / 4) [(\pi / 2 - 0) - 0] = \mathbf{9\pi / 8}$.

Part (b): Total Area of All Four Petals (2 Points)

- **Multiply by 4:**

Total Area = $4 \cdot$ (Area of one petal) = $4 \cdot (9\pi / 8) = \mathbf{9\pi / 2}$.

Part (c): Perimeter Integral for the Quadrant I Petal (2 Points)

- **Differentiate Radius:**

$dr/d\theta = 6\cos(2\theta)$.

- **Radicand for Arc Length:**

$r^2 + (dr/d\theta)^2 = 9\sin^2(2\theta) + 36\cos^2(2\theta)$.

- **Set Up Definite Integral:**

Perimeter = $\int_{0}^{\pi/2} \sqrt{[9\sin^2(2\theta) + 36\cos^2(2\theta)]} \, d\theta$ (or $3 \int_{0}^{\pi/2} \sqrt{[1 + 3\cos^2(2\theta)]} \, d\theta$).

Part (d): Radial Rate of Change at $\theta = \pi / 6$ (2 Points)

- **Evaluate $dr/d\theta$ at $\theta = \pi / 6$:**

$dr/d\theta \,|_{(\theta=\pi/6)} = 6\cos(2(\pi/6)) = 6\cos(\pi/3) = 6(1/2) = \mathbf{3}$.

- **Analyze Distance to Pole:**

The distance from the origin to the point on the curve is $|r|$.

At $\theta = \pi/6$: $r = 3\sin(\pi/3) = 3\sqrt{3} / 2 > 0$.

- **Conclusion:**

Since $r > 0$ and $dr/d\theta = 3 > 0$, the radius is growing, so the distance from the point on the curve to the pole is **increasing**.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Interval $[0, \pi/2]$: 1 pt Integral setup: 1 pt Single petal area = $9\pi / 8$: 1 pt	**3**
Part (b)	Total area = $4 \times (9\pi / 8) = 9\pi / 2$: 2 pts	**2**
Part (c)	Integral expression for arc length: 2 pts	**2**
Part (d)	$dr/d\theta \,|_{(\pi/6)} = 3$: 1 pt Conclusion that distance is increasing with reasoning: 1 pt	**2**

FRQ 8: Two Particles — Collision vs. Intersecting Paths Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Collision Analysis (3 Points)

- **Condition for Collision:**

Particles collide if and only if both $x_A(t) = x_B(t)$ and $y_A(t) = y_B(t)$ for the same time $t \geq 0$.

- **Equate x-Coordinates:**

$2t - 1 = t^2 - 4 \Rightarrow t^2 - 2t - 3 = 0 \Rightarrow (t - 3)(t + 1) = 0$.

Since $t \geq 0$, the only candidate time is $t = 3$.

- **Test y-Coordinates at t = 3:**

$y_A(3) = 3^2 - 4(3) + 7 = 9 - 12 + 7 = 4$.

$y_B(3) = 3(3) - 2 = 9 - 2 = 7$.

- **Conclusion:**

Since $y_A(3) = 4 \neq y_B(3) = 7$, the particles are at different vertical locations at $t = 3$. Therefore, the particles **never collide**.

Part (b): Rectangular Equation for Particle A (3 Points)

- **Solve for t in Terms of x:**

$x = 2t - 1 \Rightarrow 2t = x + 1 \Rightarrow t = (x + 1) / 2$.

- **Substitute into $y_A(t)$:**

$y = [(x + 1) / 2]^2 - 4[(x + 1) / 2] + 7$.

- **Expand and Simplify:**

$y = (x^2 + 2x + 1) / 4 - 2(x + 1) + 7 = (x^2 + 2x + 1) / 4 - 2x + 5$

$= (x^2 + 2x + 1 - 8x + 20) / 4 =$ **$(x^2 - 6x + 21) / 4$** (for $x \geq -1$).

Part (c): Speed Comparison at t = 3 (3 Points)

- **Velocity & Speed of Particle A at t = 3:**

$v_A(t) = \langle 2, \ 2t - 4 \rangle \Rightarrow v_A(3) = \langle 2, \ 2(3) - 4 \rangle = \langle 2, 2 \rangle$.

$\text{Speed}_A = \sqrt{[2^2 + 2^2]} = \sqrt{8} = 2\sqrt{2} \approx 2.828$.

- **Velocity & Speed of Particle B at t = 3:**

$v_B(t) = \langle 2t, \ 3 \rangle \Rightarrow v_B(3) = \langle 2(3), \ 3 \rangle = \langle 6, 3 \rangle$.

$\text{Speed}_B = \sqrt{[6^2 + 3^2]} = \sqrt{[36 + 9]} = \sqrt{45} = 3\sqrt{5} \approx 6.708$.

- **Conclusion:**

Since $3\sqrt{5} > 2\sqrt{2}$ ($\sqrt{45} > \sqrt{8}$), **Particle B is moving faster** at $t = 3$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Sets $x_A(t) = x_B(t)$ yielding $t = 3$: 1 pt Evaluates $y_A(3) = 4$ and $y_B(3) = 7$: 1 pt Correct conclusion of no collision: 1 pt	**3**
Part (b)	Solves for $t = (x + 1) / 2$: 1 pt Substitutes into $y_A(t)$: 1 pt Simplified equation $y = (x^2 - 6x + 21) / 4$: 1 pt	**3**
Part (c)	Speed of Particle A = $2\sqrt{2}$: 1 pt Speed of Particle B = $3\sqrt{5}$: 1 pt Conclusion that Particle B is faster: 1 pt	**3**

FRQ 9: Applied Vector Rate Modeling — Oceanic Vessel Navigation Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Net Velocity Vector & Speed at t = 0 (2 Points)

- **Evaluate Net Velocity at t = 0:**

$v(0) = \langle 4 + 2\sin(0), \ 3\cos(0) - 1 \rangle = \langle 4 + 0, \ 3(1) - 1 \rangle = \langle 4, 2 \rangle$ knots.

- **Compute Speed:**

$||v(0)|| = \sqrt{[4^2 + 2^2]} = \sqrt{[16 + 4]} = \sqrt{20} = \mathbf{2\sqrt{5}}$ knots.

Part (b): Vessel Position at t = 2π (3 Points)

- **Integrate x'(t) from 0 to 2π:**

$x(2\pi) = 0 + \int_0^{2\pi} (4 + 2\sin(t/4))\, dt = [4t - 8\cos(t/4)]_0^{(2\pi)}$

$= [4(2\pi) - 8\cos(2\pi/4)] - [0 - 8\cos(0)]$

$= [8\pi - 8\cos(\pi/2)] - [-8(1)] = [8\pi - 0] + 8 = 8\pi + 8.$

- **Integrate y'(t) from 0 to 2π:**

$y(2\pi) = 0 + \int_0^{2\pi} (3\cos(t/2) - 1)\, dt = [6\sin(t/2) - t]_0^{(2\pi)}$

$= [6\sin(\pi) - 2\pi] - [6\sin(0) - 0] = 0 - 2\pi - 0 = -2\pi.$

- **Position:** $(8\pi + 8, -2\pi)$.

Part (c): Acceleration Vector at t = 2π (2 Points)

- **Differentiate Velocity Components:**

$a(t) = v'(t) = \langle d/dt(4 + 2\sin(t/4)), \ d/dt(3\cos(t/2) - 1) \rangle = \langle (1/2)\cos(t/4), \ -(3/2)\sin(t/2) \rangle.$

- **Evaluate at t = 2π:**

$a(2\pi) = \langle (1/2)\cos(\pi/2), \ -(3/2)\sin(\pi) \rangle = \langle (1/2)(0), \ -(3/2)(0) \rangle = \mathbf{\langle 0, 0 \rangle}.$

Part (d): Total Distance Traveled Integral (2 Points)

- **Form the Arc Length Integral:**

Distance $= \int_0^{12} \sqrt{[(4 + 2\sin(t/4))^2 + (3\cos(t/2) - 1)^2]}\, dt.$

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Velocity $v(0) = \langle 4, 2 \rangle$: 1 pt Speed $= 2\sqrt{5}$ knots: 1 pt	**2**
Part (b)	$x(2\pi)$ integration: 1 pt $y(2\pi)$ integration: 1 pt Position $(8\pi + 8, -2\pi)$: 1 pt	**3**
Part (c)	Derivative expressions: 1 pt Acceleration vector $a(2\pi) = \langle 0, 0 \rangle$: 1 pt	**2**
Part (d)	Integral expression over $[0, 12]$: 2 pts	**2**

FRQ 10: Comprehensive Unit 9 Capstone Synthesis Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (a): Position and Radial Velocity at $\theta = \pi / 3$ (2 Points)

- **Evaluate Radius at $\theta = \pi / 3$:**

$r(\pi/3) = 2 + 4\cos(\pi/3) = 2 + 4(1/2) = 4$.

- **Compute Cartesian Coordinates:**

$x = r\cos(\pi/3) = 4(1/2) = 2$.

$y = r\sin(\pi/3) = 4(\sqrt{3}/2) = 2\sqrt{3}$.

Position is **$(2, 2\sqrt{3})$**.

- **Compute dr/dt using the Chain Rule:**

$dr/d\theta = -4\sin\theta$.

$dr/dt = (dr/d\theta) \cdot (d\theta/dt) = (-4\sin(\pi/3)) \cdot 2 = -8(\sqrt{3}/2) = \mathbf{-4\sqrt{3}}$.

Part (b): Velocity Vector and Speed at $\theta = \pi / 3$ (3 Points)

- **Horizontal Derivative dx/dt:**

$x(\theta) = (2 + 4\cos\theta)\cos\theta = 2\cos\theta + 4\cos^2\theta$.

$dx/d\theta = -2\sin\theta - 8\cos\theta\sin\theta = -2\sin\theta - 4\sin(2\theta)$.

At $\theta = \pi/3$: $dx/d\theta = -2(\sqrt{3}/2) - 4(\sqrt{3}/2) = -\sqrt{3} - 2\sqrt{3} = -3\sqrt{3}$.

$dx/dt = (dx/d\theta) \cdot (d\theta/dt) = (-3\sqrt{3}) \cdot 2 = \mathbf{-6\sqrt{3}}$.

- **Vertical Derivative dy/dt:**

$y(\theta) = (2 + 4\cos\theta)\sin\theta = 2\sin\theta + 4\cos\theta\sin\theta = 2\sin\theta + 2\sin(2\theta)$.

$dy/d\theta = 2\cos\theta + 4\cos(2\theta)$.

At $\theta = \pi/3$: $dy/d\theta = 2(1/2) + 4\cos(2\pi/3) = 1 + 4(-1/2) = 1 - 2 = -1$.

$dy/dt = (dy/d\theta) \cdot (d\theta/dt) = (-1) \cdot 2 = \mathbf{-2}$.

- **Velocity Vector:**

$v = \langle -6\sqrt{3},   -2 \rangle$.

- **Speed:**

$||v|| = \sqrt{[(-6\sqrt{3})^2 + (-2)^2]} = \sqrt{[108 + 4]} = \sqrt{112} = \mathbf{4\sqrt{7}}$.

Part (c): Slope of Tangent Line at $\theta = \pi / 3$ (2 Points)

- **Evaluate dy/dx:**

$dy/dx = (dy/dt) / (dx/dt) = (-2) / (-6\sqrt{3}) = 1 / (3\sqrt{3}) = \mathbf{\sqrt{3} / 9}$.

Part (d): Area of Inner Loop Integral (2 Points)

- **Find Roots for the Inner Loop:**

$r = 0 \Longrightarrow 2 + 4\cos\theta = 0 \Longrightarrow \cos\theta = -1/2 \Longrightarrow \theta = 2\pi/3$ and $\theta = 4\pi/3$.

- **Set Up Area Integral:**

Area $= (1/2)\int^{2\pi/3}_{4\pi/3} (2 + 4\cos\theta)^2\, d\theta$ (or $\int^{2\pi/3}_{\pi} (2 + 4\cos\theta)^2\, d\theta$).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (a)	Cartesian position $(2, 2\sqrt{3})$: 1 pt Rate of change $dr/dt = -4\sqrt{3}$: 1 pt	**2**
Part (b)	$dx/dt = -6\sqrt{3}$ and $dy/dt = -2$: 2 pts Speed $= 4\sqrt{7}$: 1 pt	**3**
Part (c)	Tangent slope $dy/dx = \sqrt{3}/9$: 2 pts	**2**
Part (d)	Area integral for inner loop: 2 pts	**2**

Essential Unit 9 Rules

Core formulas, parametric derivatives, polar area theorems, and vector kinematics required for AP Calculus BC.

- **Parametric Derivatives & Tangent Slopes:** • First Derivative: $dy/dx = (dy/dt) / (dx/dt)$ provided $dx/dt \neq 0$. • Horizontal Tangent: $dy/dt = 0$ and $dx/dt \neq 0$. Vertical Tangent: $dx/dt = 0$ and $dy/dt \neq 0$. • Second Derivative: $d^2y/dx^2 = [\, d/dt(dy/dx) \,] / [\, dx/dt \,]$. Never divide d^2y/dt^2 by d^2x/dt^2!
- **Parametric Arc Length & Planar Distance:** • Total Distance Traveled = $\int_{t_1}^{t_2} \sqrt{[\, (x'(t))^2 + (y'(t))^2 \,]}\, dt = \int_{t_1}^{t_2} ||v(t)||\, dt$.
- **Vector-Valued Functions & Planar Kinematics:** • Position: $r(t) = \langle x(t), y(t) \rangle$. Velocity: $v(t) = r'(t) = \langle x'(t), y'(t) \rangle$. Acceleration: $a(t) = v'(t) = \langle x''(t), y''(t) \rangle$. • Speed: $||v(t)|| = \sqrt{[\, (x'(t))^2 + (y'(t))^2 \,]}$. Direction Angle: $\tan\theta = y'(t) / x'(t)$. • Rate of Change of Speed: $d/dt\, ||v(t)|| = [\, v(t) \cdot a(t) \,] / ||v(t)|| = [\, x'x'' + y'y'' \,] / \sqrt{[\, (x')^2 + (y')^2 \,]}$. • Speed is increasing when $v \cdot a > 0$; decreasing when $v \cdot a < 0$. Particle is at rest when $v(t) = \langle 0, 0 \rangle$.
- **Polar Coordinates & Conversions:** • Coordinate relations: $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$, $\tan\theta = y / x$. • Slope of Polar Tangent Line: $dy/dx = (dy/d\theta) / (dx/d\theta) = [\, (dr/d\theta)\sin\theta + r\cos\theta \,] / [\, (dr/d\theta)\cos\theta - r\sin\theta \,]$. • Tangents at the Pole: If $r(\theta_0) = 0$ and $r'(\theta_0) \neq 0$, the line tangent to the curve at the pole is $\theta = \theta_0$.
- **Area of Polar Regions:** • Single Polar Curve: Area = $(1/2) \int_{\alpha}^{\beta} [r(\theta)]^2\, d\theta$. • Area Between Two Polar Curves: Area = $(1/2) \int_{\alpha}^{\beta} [\, (R(\theta))^2 - (r(\theta))^2 \,]\, d\theta$, where $R(\theta)$ is outer radius and $r(\theta)$ is inner radius. • Rate of Change of Swept Area: $dA/d\theta = (1/2)[r(\theta)]^2$ and $dA/dt = (1/2)[r(\theta)]^2\, (d\theta/dt)$.
- **Length of a Polar Curve (Polar Arc Length):** • Arc Length $L = \int_{\alpha}^{\beta} \sqrt{[\, [r(\theta)]^2 + [dr/d\theta]^2 \,]}\, d\theta$.

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Thank You, Students!

Mastery Comes Through Deliberate Practice

Congratulations on completing this comprehensive 100-Question Priority-Starred Master Repository for Unit 9! Parametric equations, polar coordinates, and vector-valued functions are the defining topics that elevate AP Calculus BC into multi-dimensional analysis and planar kinematics. Review your steps, reflect on the distractor rationales, and approach your AP Calculus BC examination with absolute mastery and confidence!

TARGET SCORE: 5 · AP CALCULUS BC